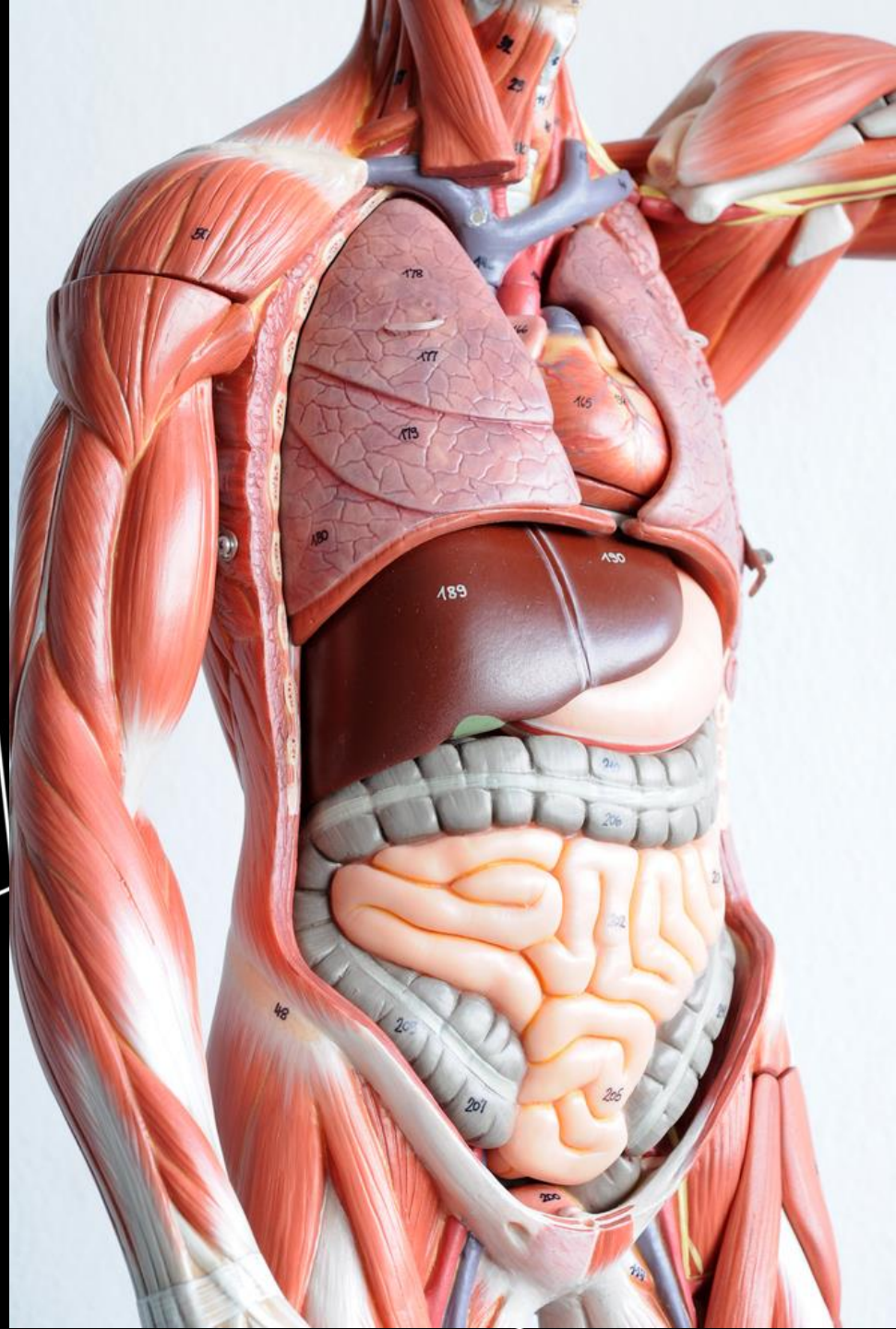
DRS - Liver Anatomy and function

Lecturer: Dr Li Yueming

Acknowledgement: Dr. Rogelio M. Levardo Jr.
Prof. Trish Lalor

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Learning Outcomes

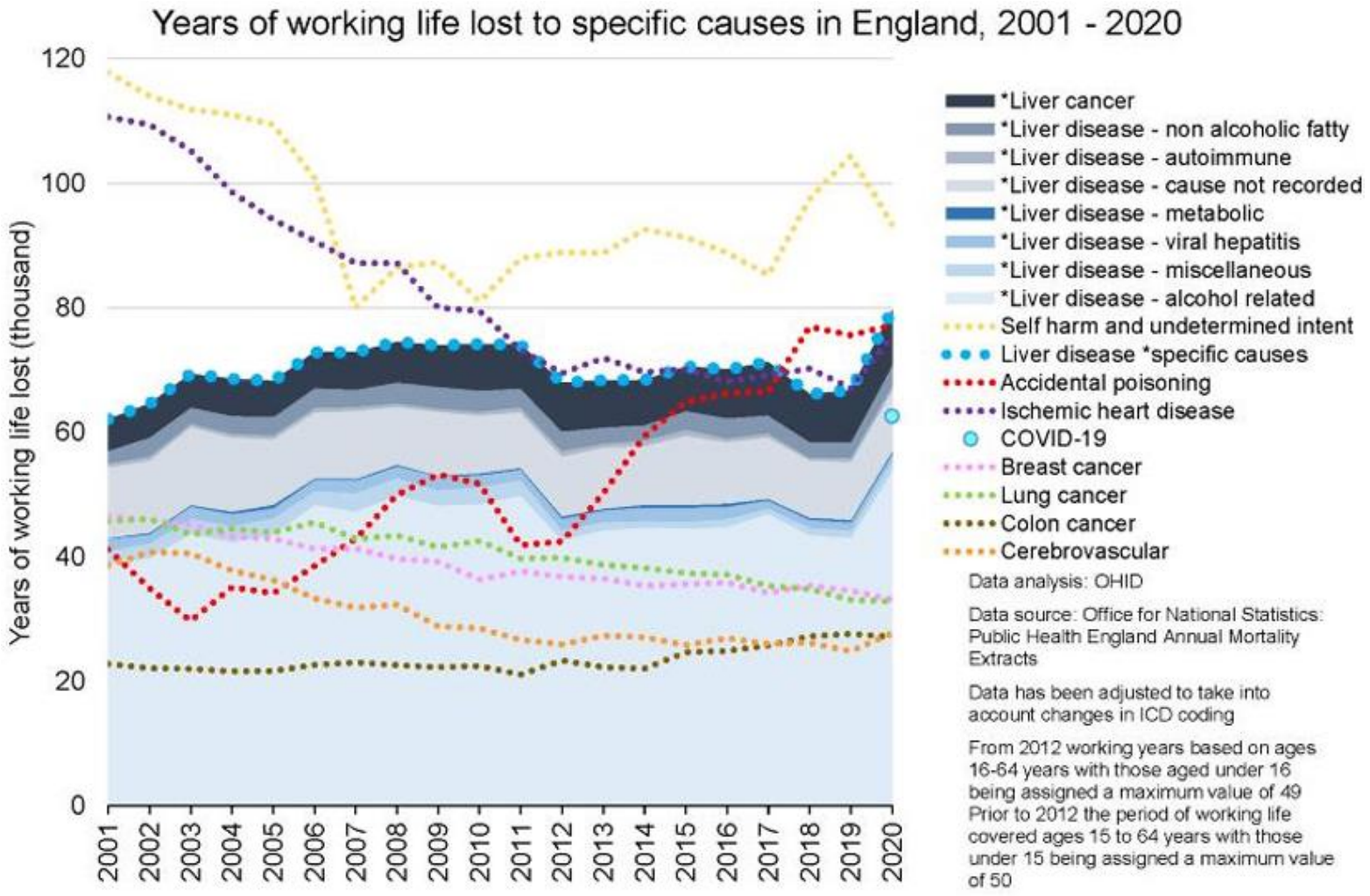
By the end of this lecture, you should be able to:

- Describe the gross and microscopic anatomy of the liver.
- Identify the major hepatic cell types.
- Explain the major functions of the liver.
- Recognise the multiple causes of liver disease.

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Why Is Liver Biology Important?

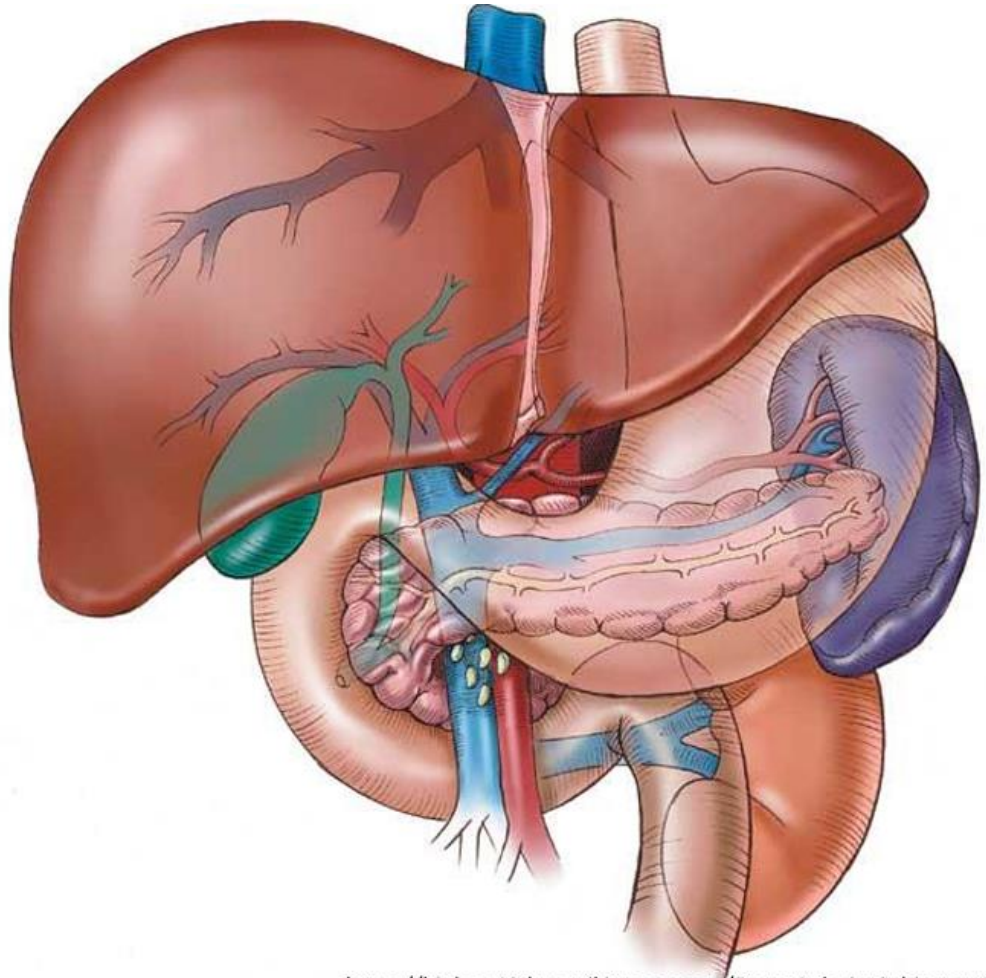
- Liver disease is an important cause of death.
- Liver disease and liver cancer caused 2.5% of deaths in England in 2020.
- Almost half of these deaths occurred in people aged 15–64.
- Liver disease can affect people of working age.



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Overview of Today's Lecture

1. **Liver anatomy**
2. **Liver cells**
3. **Liver functions**
4. **Liver diseases**



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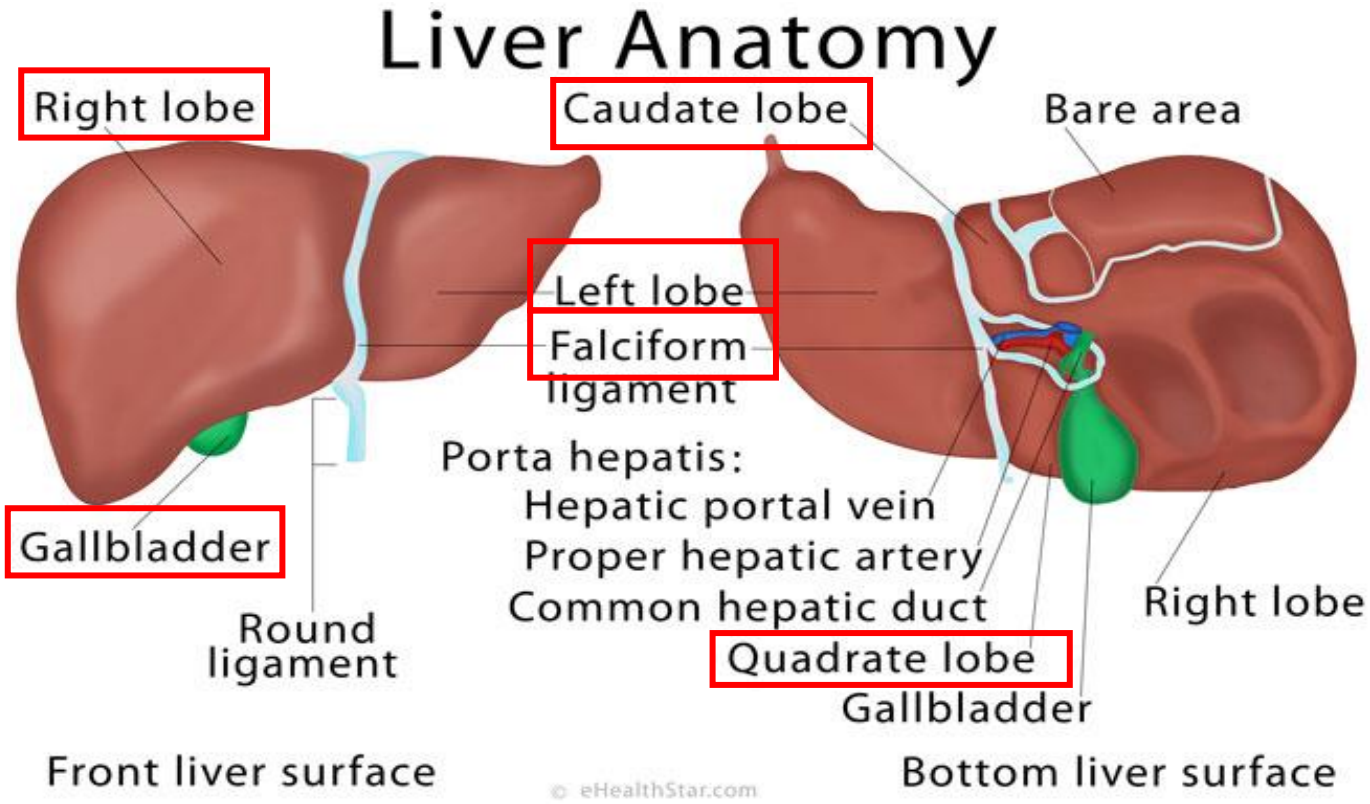
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Part 1 – Liver anatomy

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Gross Anatomy of the Liver

- The largest solid organ in the body
- Approximately 1/50 of body weight
- Located in the right upper quadrant (RUQ)
- Covered by Glisson's capsule
- Right and left lobes
- Falciform ligament separates the lobes

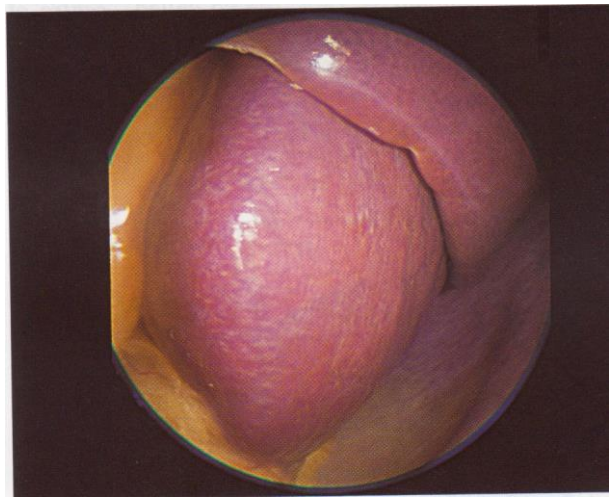


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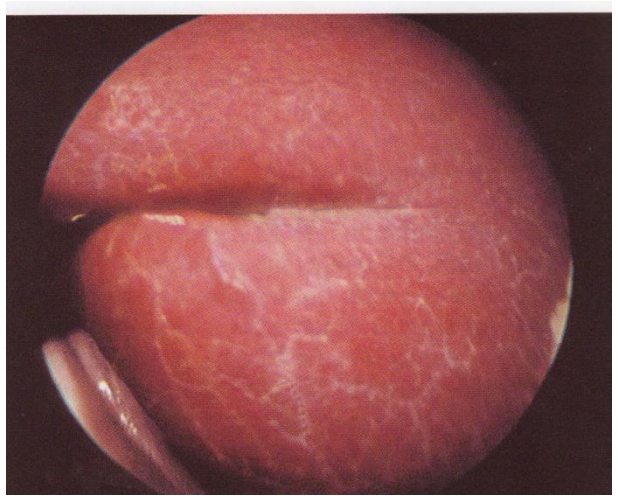
Anatomical Variations of the Liver

- Genetic variations: accessory lobes
- Internal factors: thrombosis, fibrosis and atrophy
- External factors: pressure from surrounding structures
- Examples: Riedel's lobe, accessory lobe, clefts and fissures
- Lobular atrophy may follow impaired blood supply or bile drainage

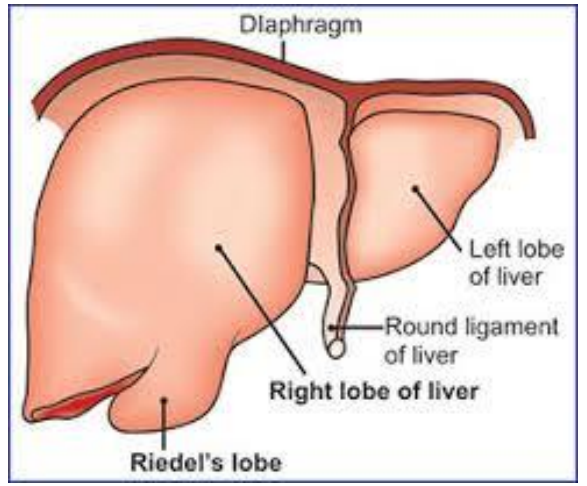
Accessory lobe



Simian Cleft



Riedel's Cleft

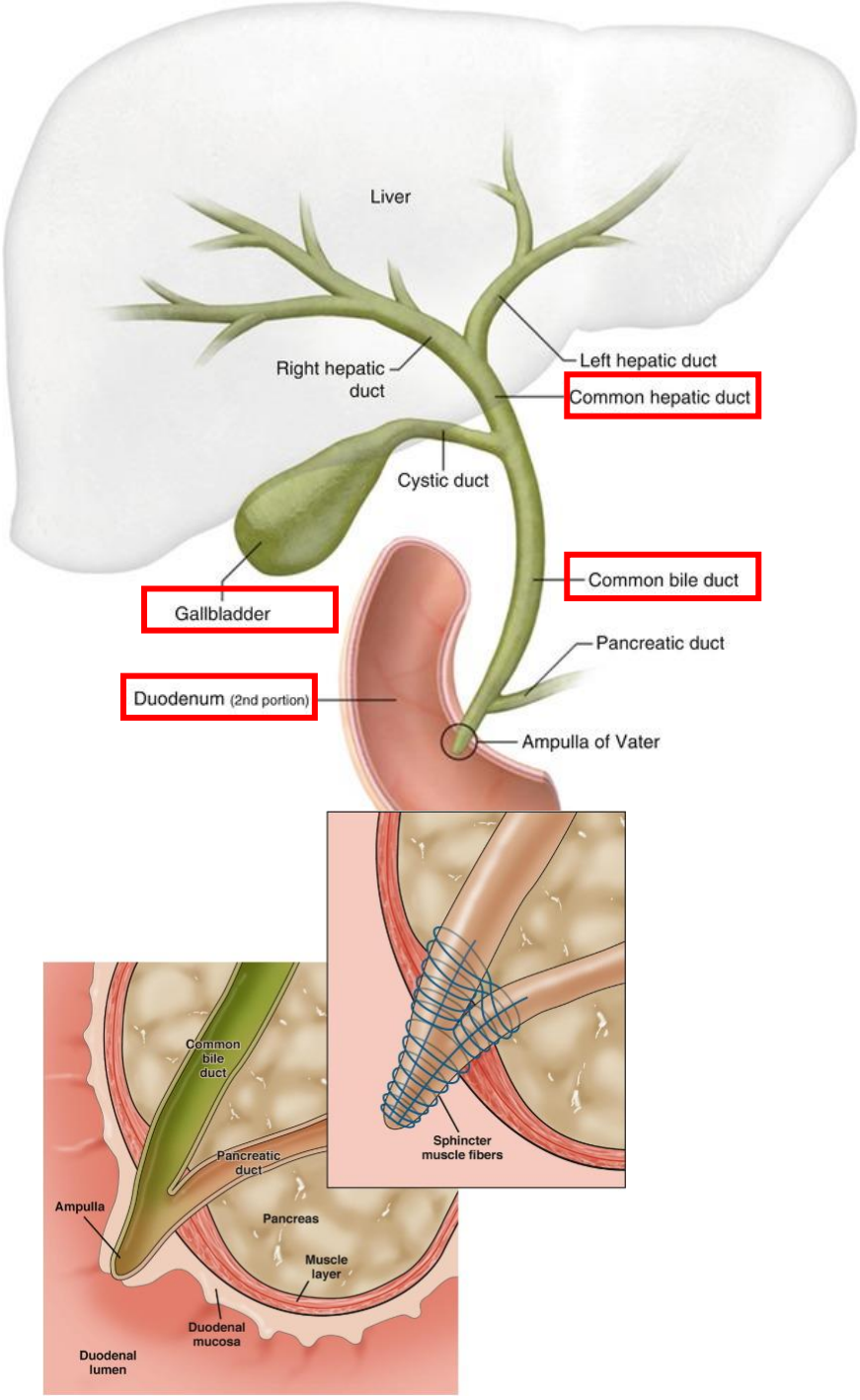


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The Gallbladder and Bile Drainage

- Gallbladder: stores, concentrates and releases bile
- Bile travels through the hepatic ducts
- Common bile duct delivers bile to the duodenum
- Sphincter of Oddi controls bile release

Liver → Hepatic ducts → Gallbladder → Common bile duct → Duodenum



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Functional Segments of the Liver

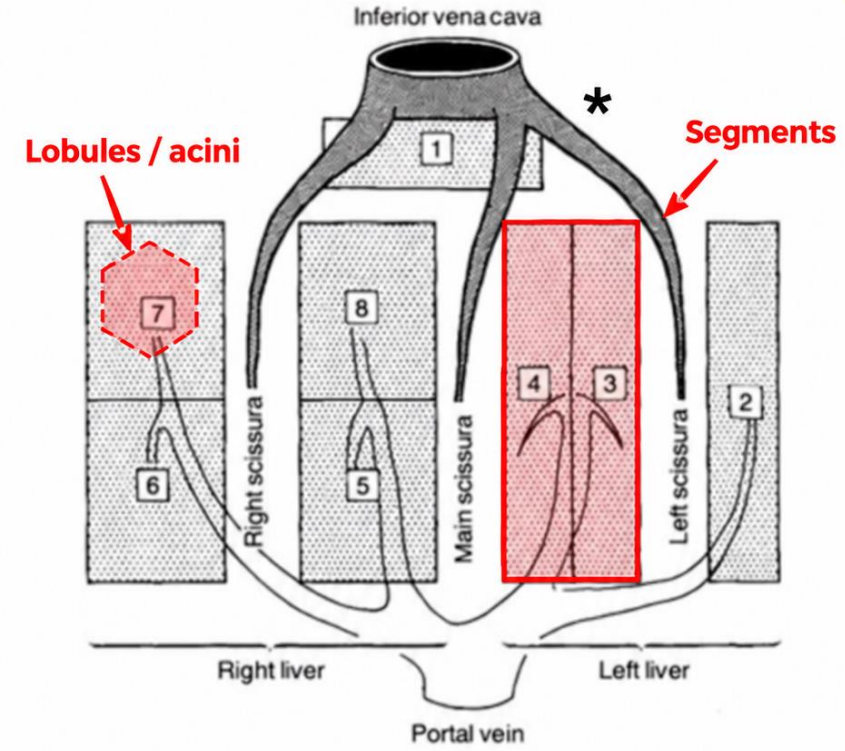
The liver can be divided into functional segments…..

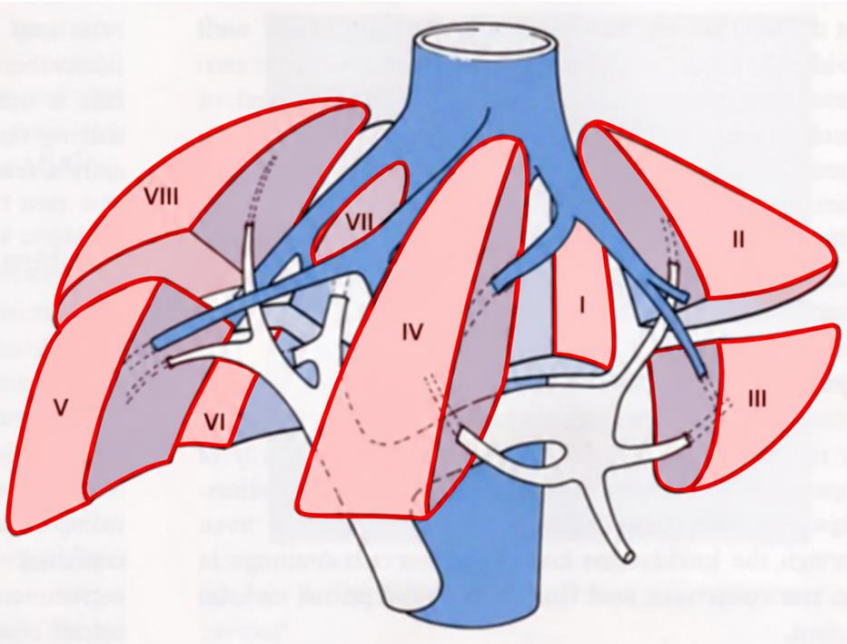
- Based upon blood supply and bile drainage
- Each segment has an independent system
- Important for maintaining functional capacity

- Within each segment, the tissue can be divided into lobule/ acinus – the functional units of the liver, composed of :

✓ plates of hepatocytes
✓ sinusoidal channels
✓ Inlet and exit blood vessels
✓ bile canaliculi

Budd-Chiari syndrome: caudate lobe hypertrophied

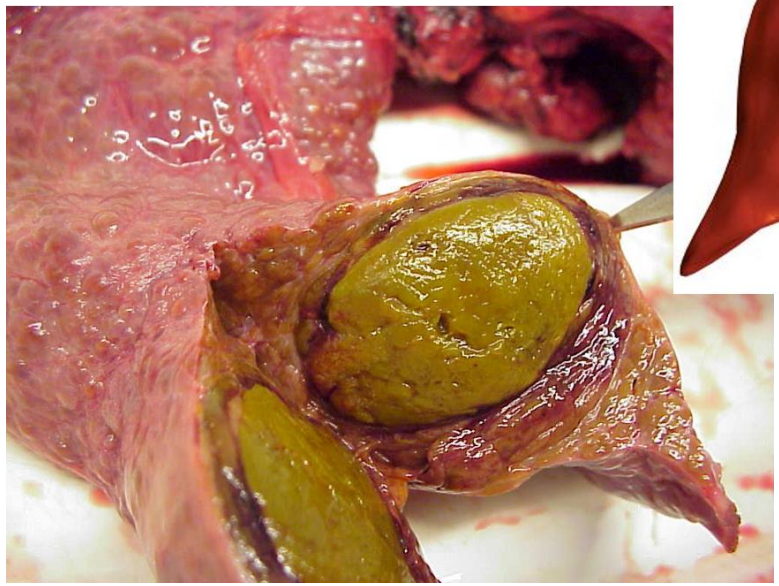


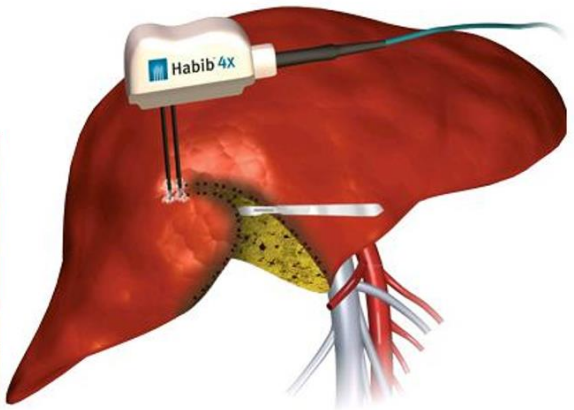


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Liver Regeneration

- The liver has a remarkable capacity to regenerate.
- Diseased tissue can be surgically removed.
- Remaining liver tissue can regenerate within weeks.
- Important for liver surgery and transplantation

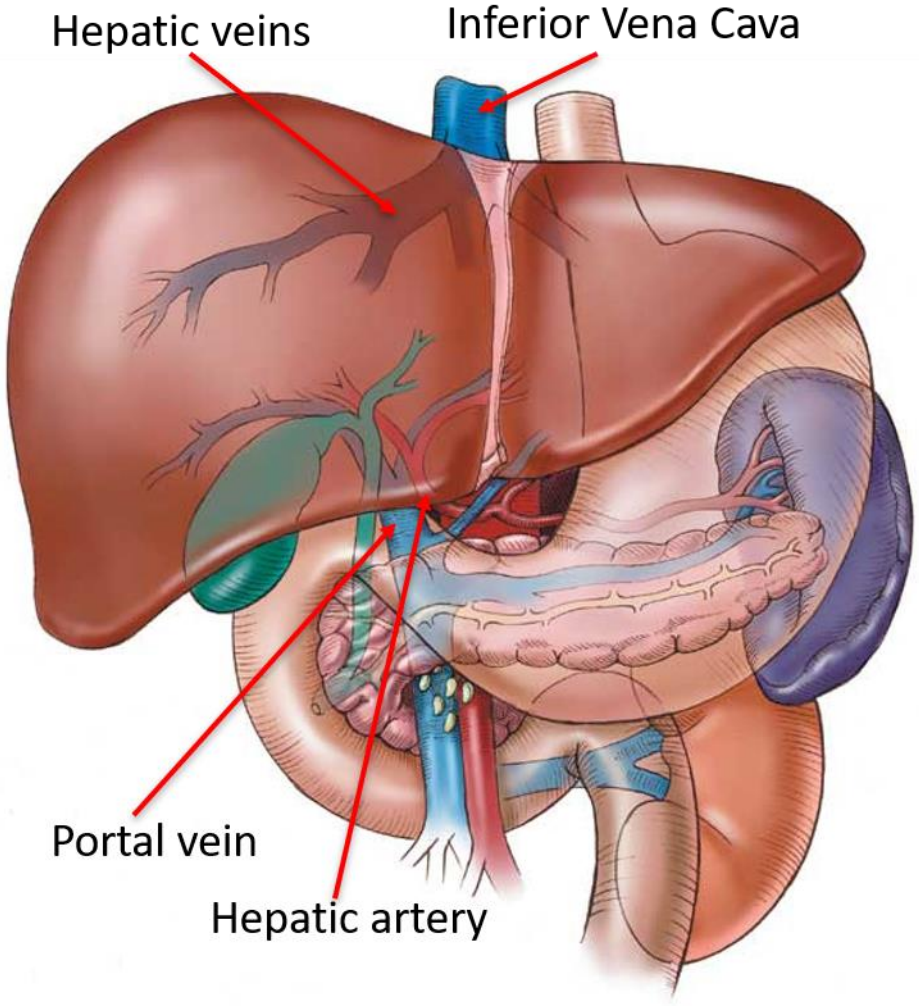




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Liver regeneration helps restore functional liver tissue

The Liver Has a Dual Blood Supply



- Portal vein – deoxygenated, nutrient-rich blood from gut (75%)
- Hepatic artery – oxygenated blood from abdominal aorta (25%)

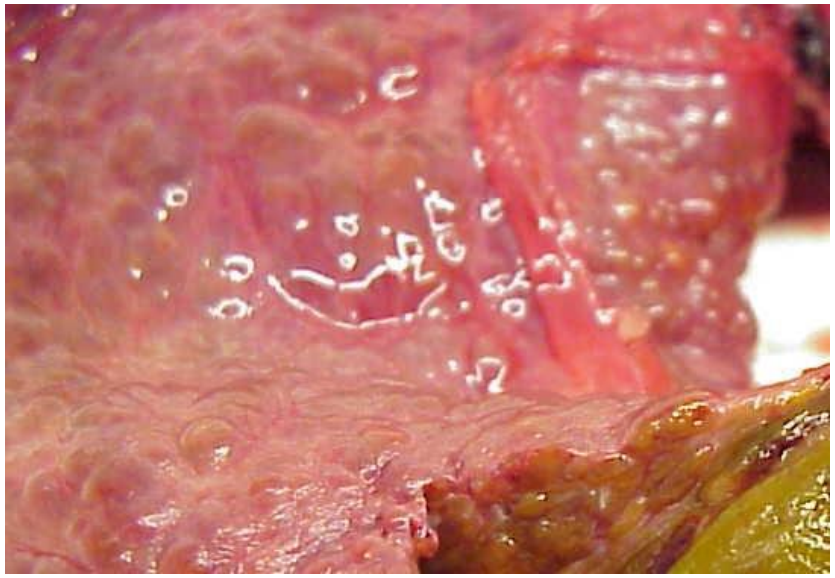
- Estimated 25% of cardiac output enters liver (1.3L+/min).

- Blood content is up to 30% of liver weight
- Liver contains up to 15% of total blood content at a time.

- Blood enters, mixes in the sinusoids (capillary network) and drains via hepatic veins into the inferior vena cava (IVC) then the right atrium

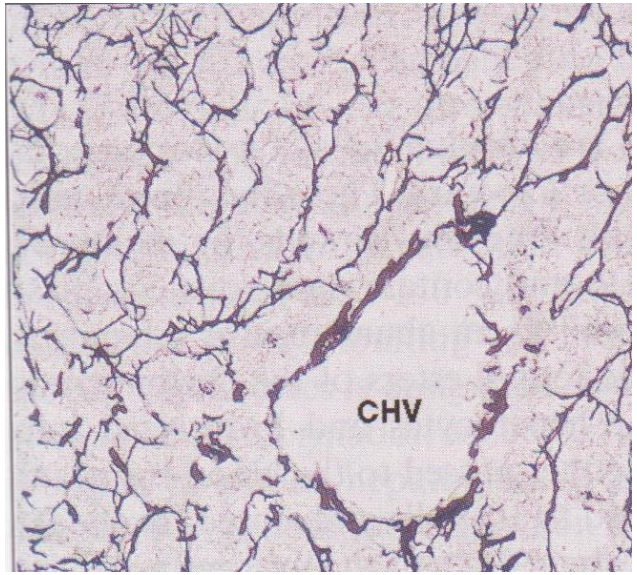
Liver Stroma and Extracellular Matrix

Liver Supportive Framework (Stroma)



- Connective Tissue – Glisson's capsule
  - ✓ perivascular tissue around blood vessels
  - ✓ portal tract / triad : contains one branch each of a hepatic artery, portal vein, and bile duct.
  - ✓ reticular network of fibres

- Composed of ECM materials – dynamic and complex macromolecules
  - ✓ mainly collagens (I, II, III, IV, V and VI) produced by stellate cells
    - Half-life 30 days, hydroxyproline produced as metabolite from the breakdown by matrix metalloproteinases (MMPs)
  - ✓ Glycoproteins link cells to collagen/ECM
  - ✓ Integrins for bidirectional communication between cells



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Liver Lobule versus Hepatic Acinus

Structural organisation **ACINUS** **LOBULE** Useful in pathophysiology

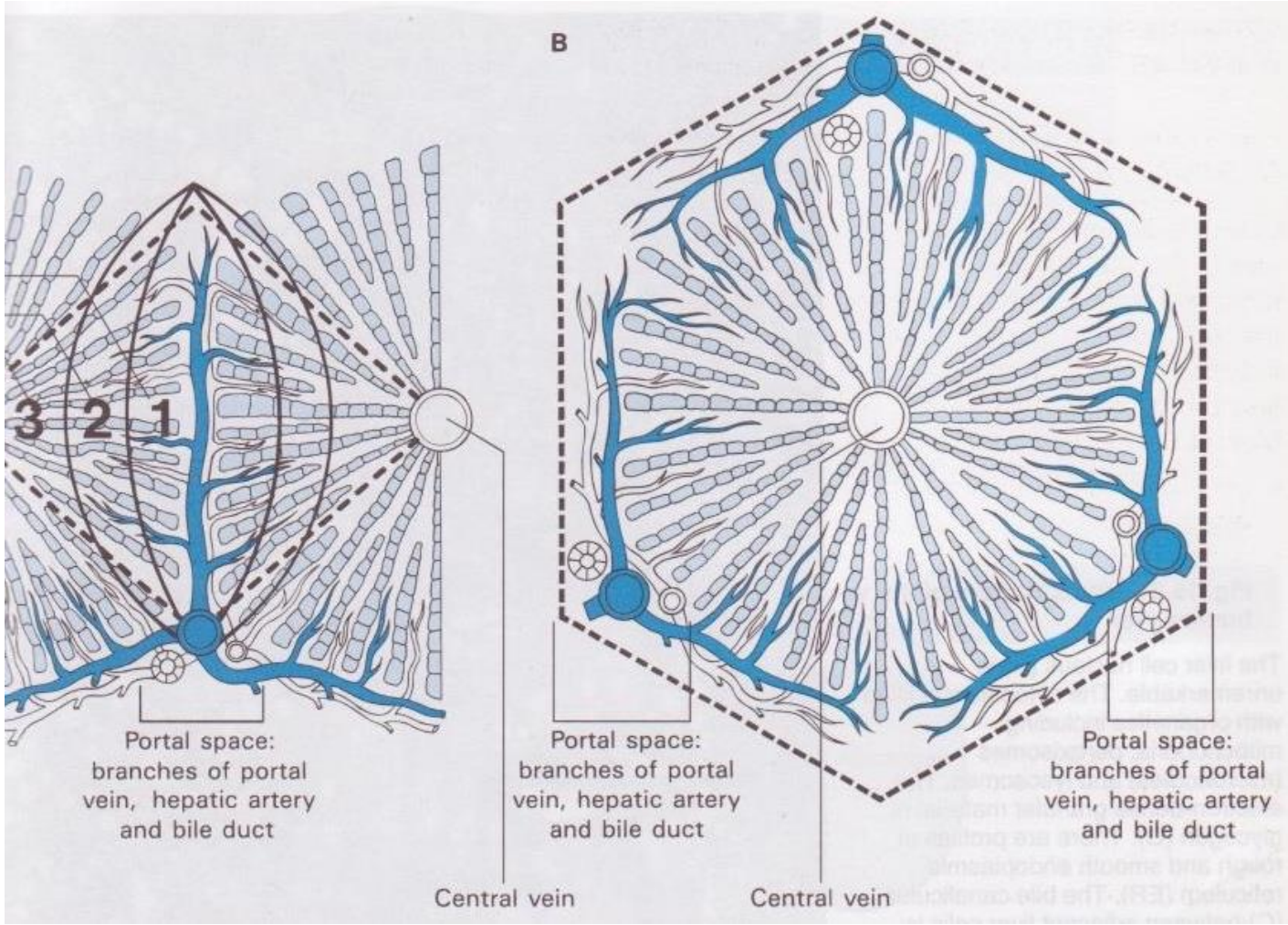
Centred on portal tracts **Centred on central veins**

Portal spaces (blue): nutrient-rich blood

Arteries: oxygen-rich blood

Autoimmune injury: seen closer to the portal spaces, bile ducts

Closer to the central vein: less oxygen and nutrients; see impact of toxic injury 1st



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Microscopic Anatomy: Portal Triad and Sinusoids

Portal triad

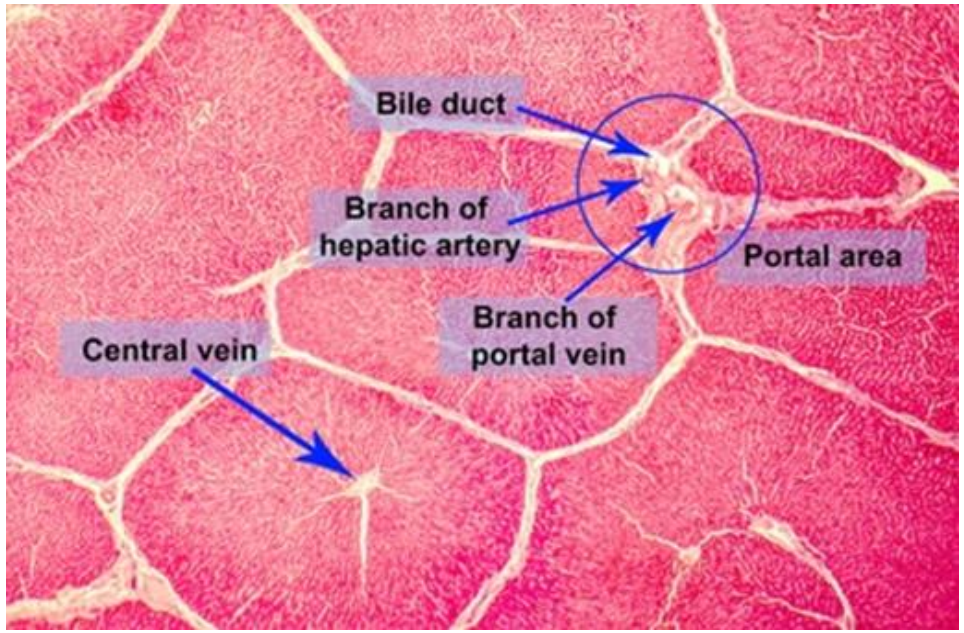
- Branch of portal vein
- Branch of hepatic artery
- Bile duct

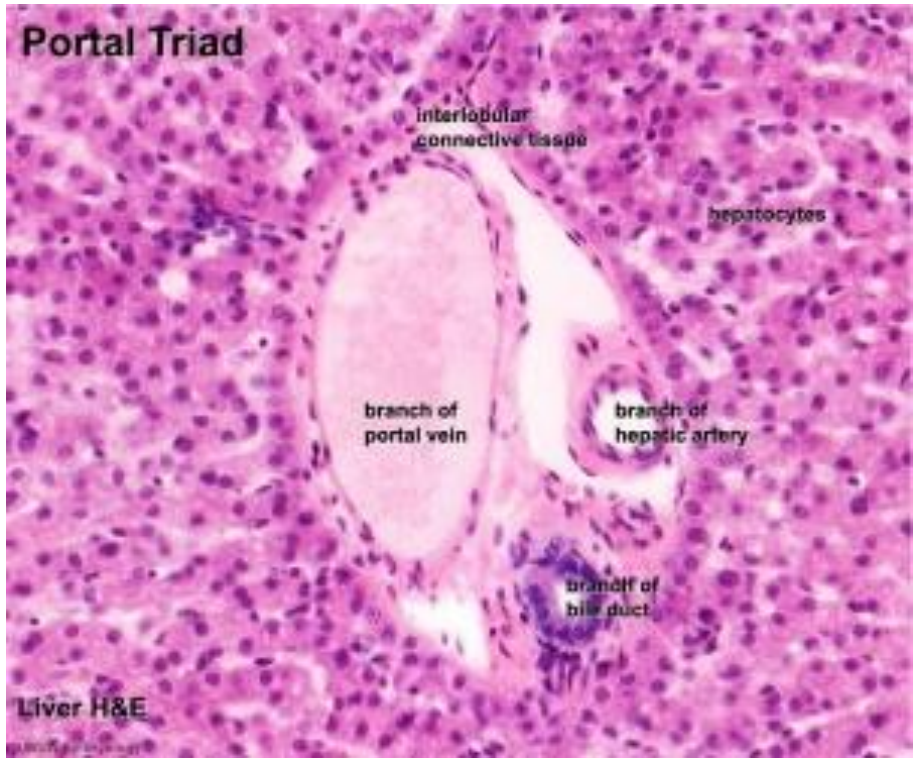
Other structures

Hepatocyte plates

Sinusoids

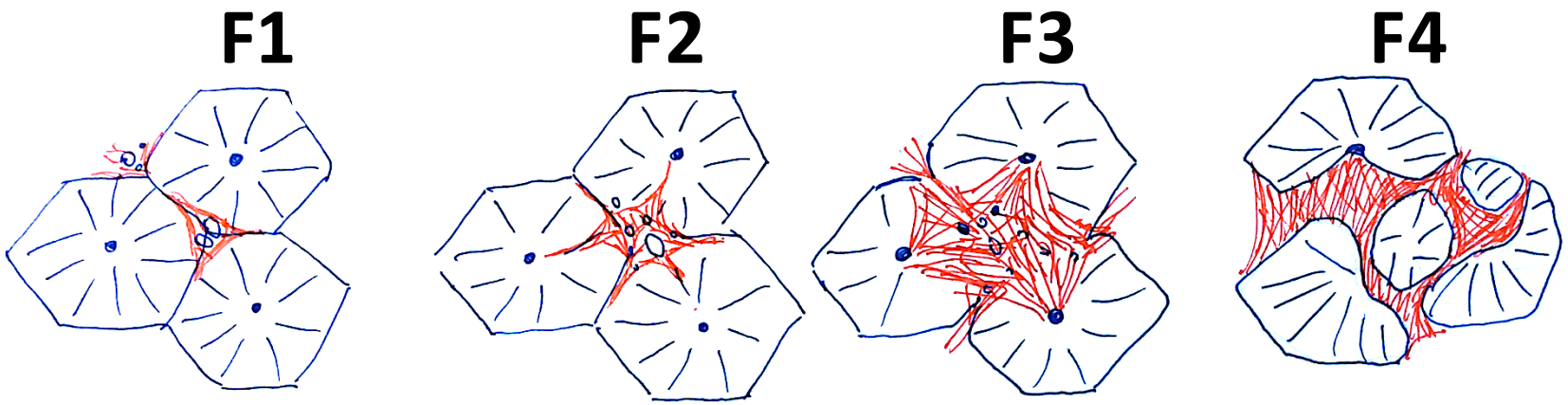
Central vein





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Fibrosis Disrupts Liver Architecture



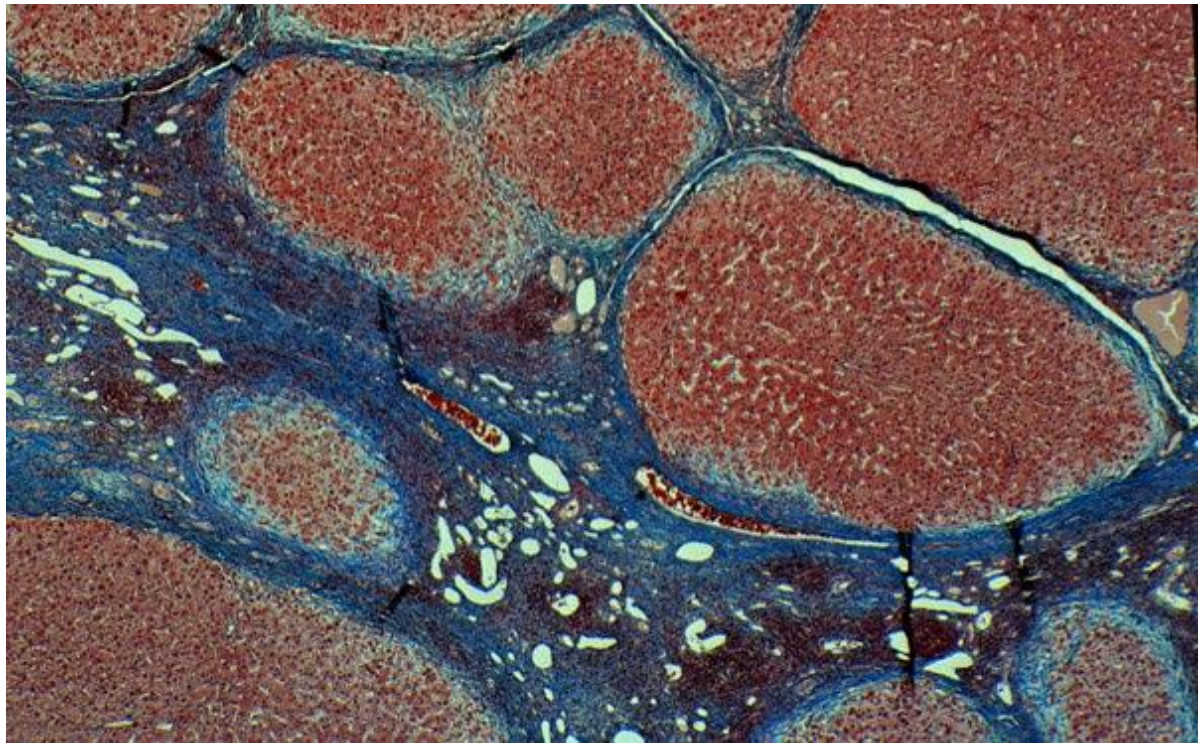
F1 Minimal fibrosis

F2 Moderate fibrosis

F3 Severe fibrosis

F4 Cirrhosis

Progressive fibrosis changes the normal organisation of liver lobules.



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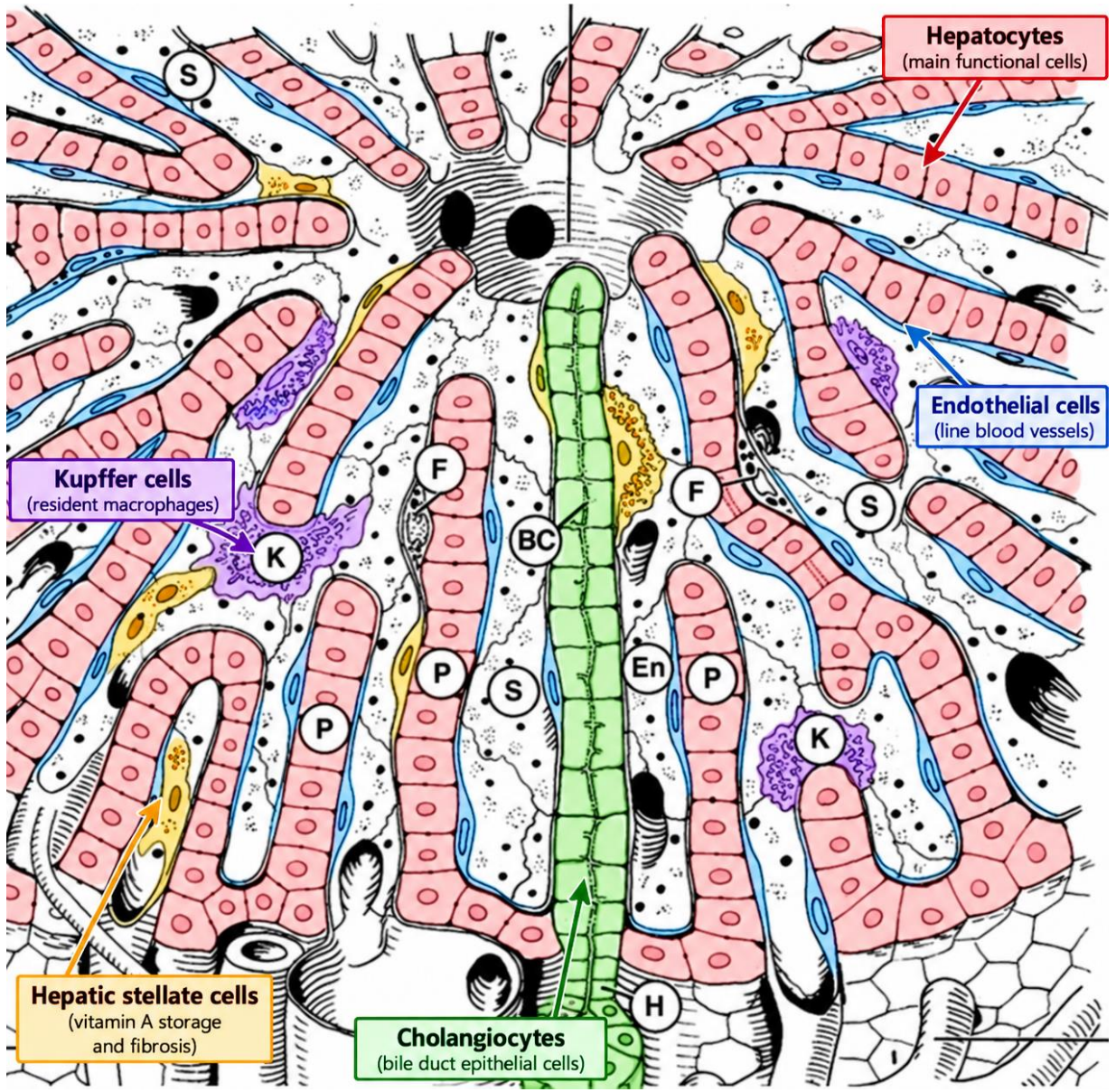
Part 2 – Liver cells

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Major Cell Types in the Liver

The liver contains several specialised cell types:

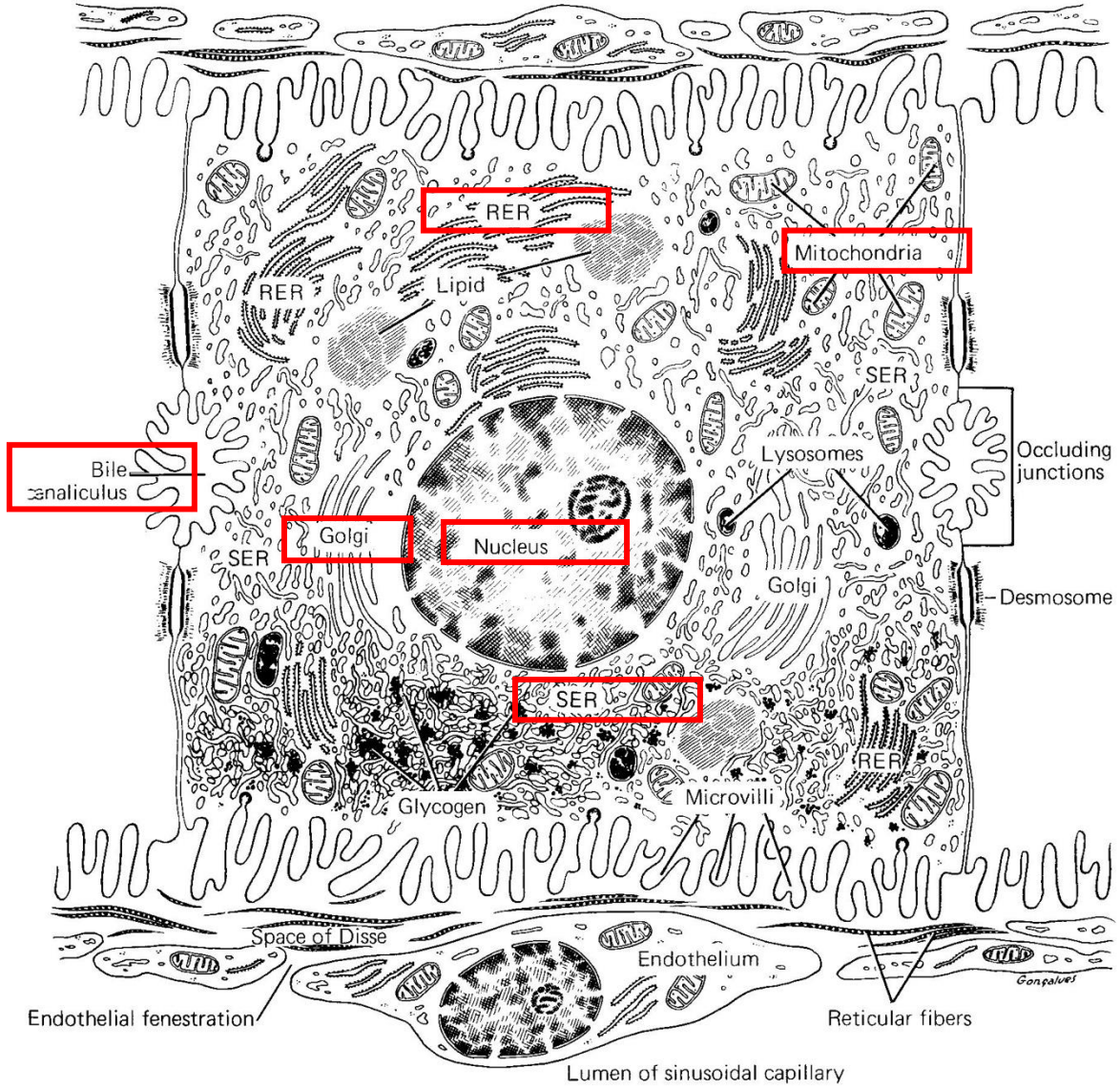
- Hepatocytes — main functional cells
- Cholangiocytes — bile duct epithelial cells
- Endothelial cells — line blood vessels
- Kupffer cells — resident macrophages
- Hepatic stellate cells — vitamin A storage and fibrosis



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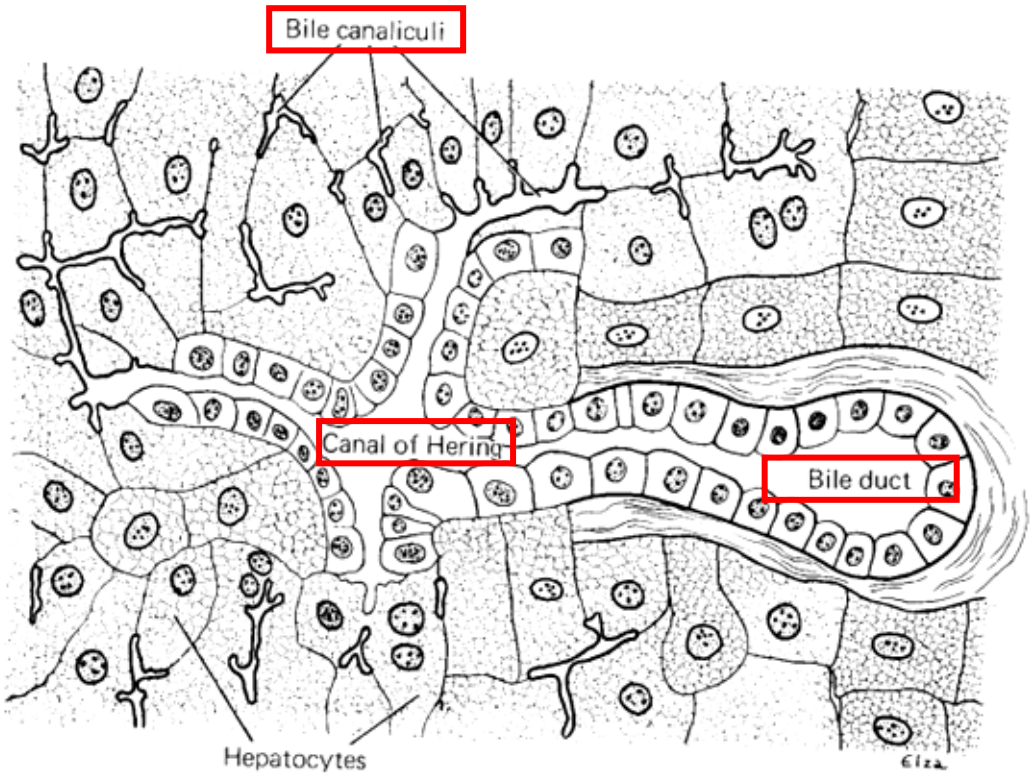
Hepatocytes: The Main Functional Cells

- 60–65% of liver tissue
- Approximately 100 billion cells
- Polarised, polyhedral epithelial cells
- Approximately 20–30 µm
- Low mitotic index
- Main functional cells of the liver



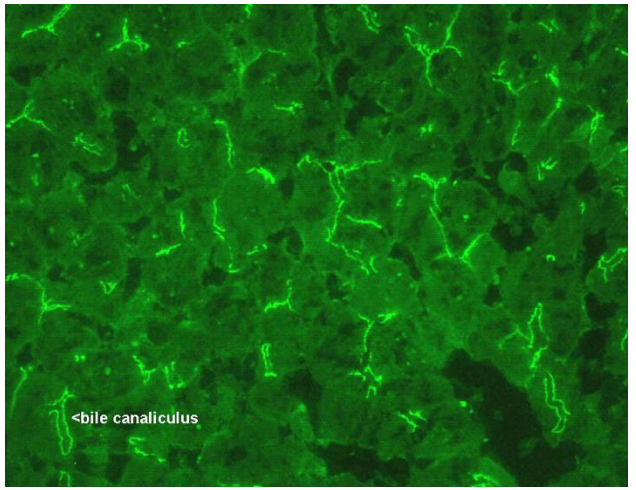
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Cholangiocytes: Bile Duct Epithelial Cells



Cholangiocytes = biliary epithelial cells

- 1–3% of liver tissue
- Polarised cuboidal or columnar epithelial cells
- Line the bile ducts
- Surrounded by a dense basement membrane
- Transport, modify and release bile



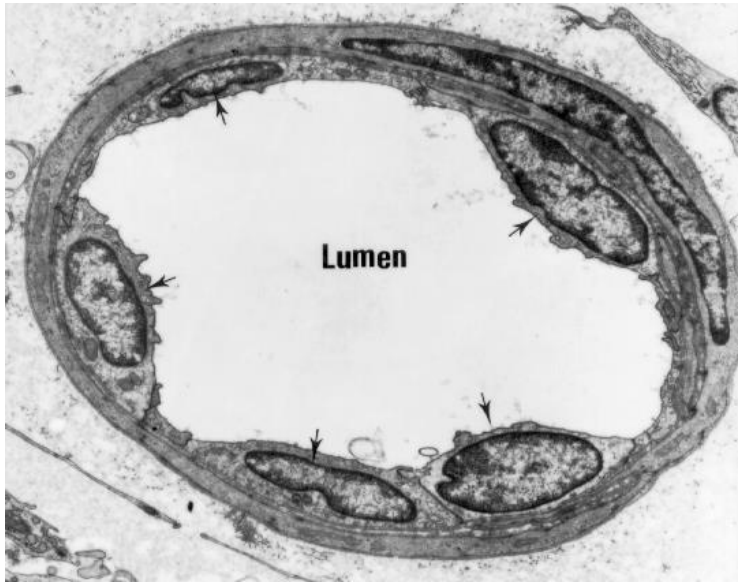
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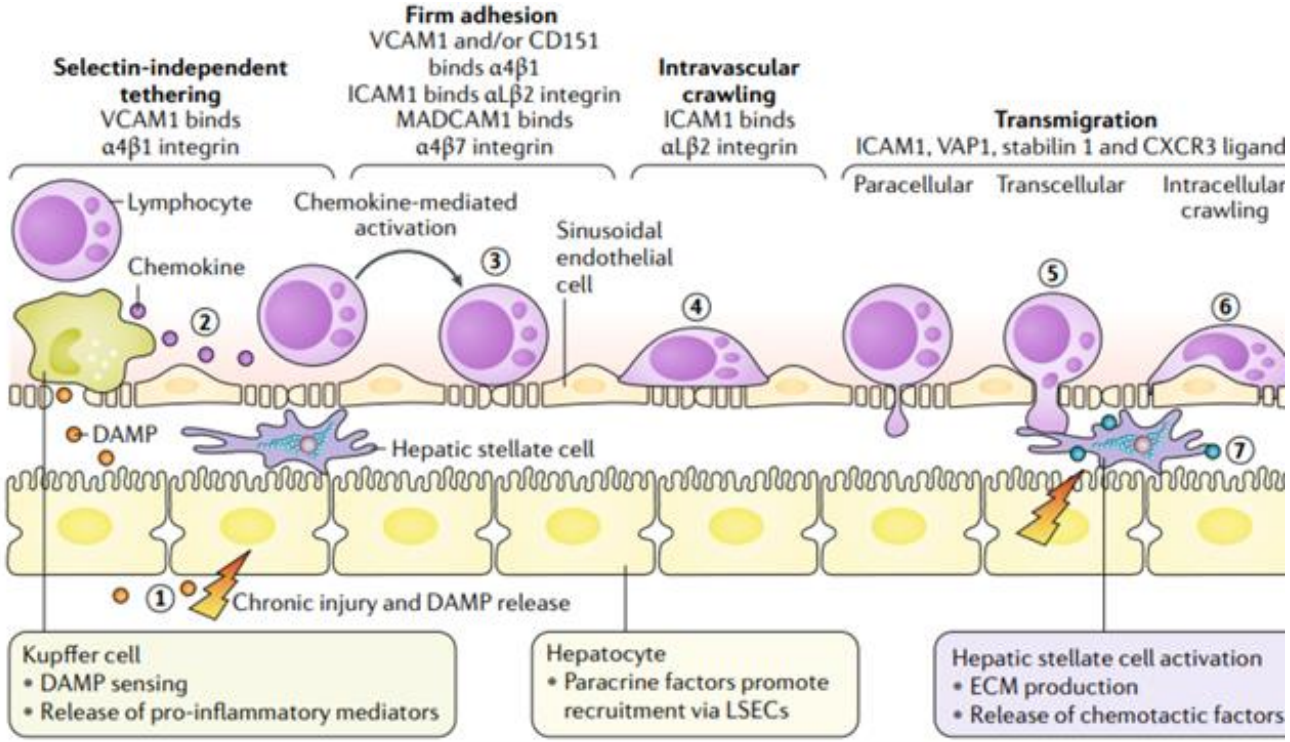
Hepatic Endothelial Cells

Location: Hepatic blood vessels and sinusoids

Functions:

- Form a barrier between blood and liver tissue
- Filter fluids
- Regulate blood coagulation
- Selective uptake of solutes and particles
- Remove waste products
- Regulate immune cell movement



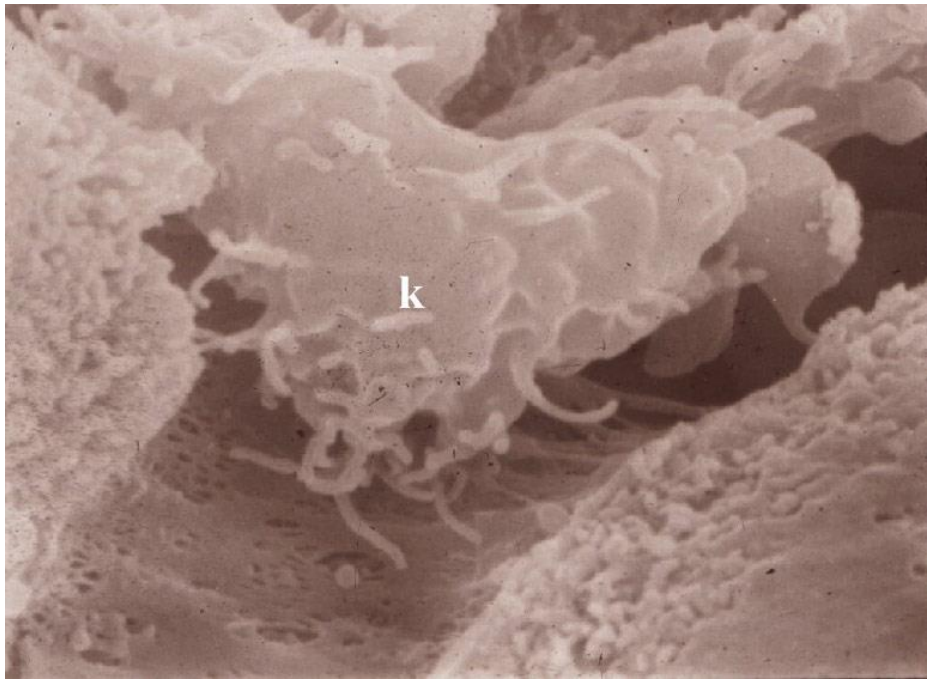


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Kupffer Cells: Liver Macrophages

Kupffer cells = resident hepatic macrophages

- Located within the sinusoids
- Phagocytosis and endocytosis
- Removal of endotoxins
- Regulation of microcirculation
- Cytokine production
- Antigen presentation and immune responses

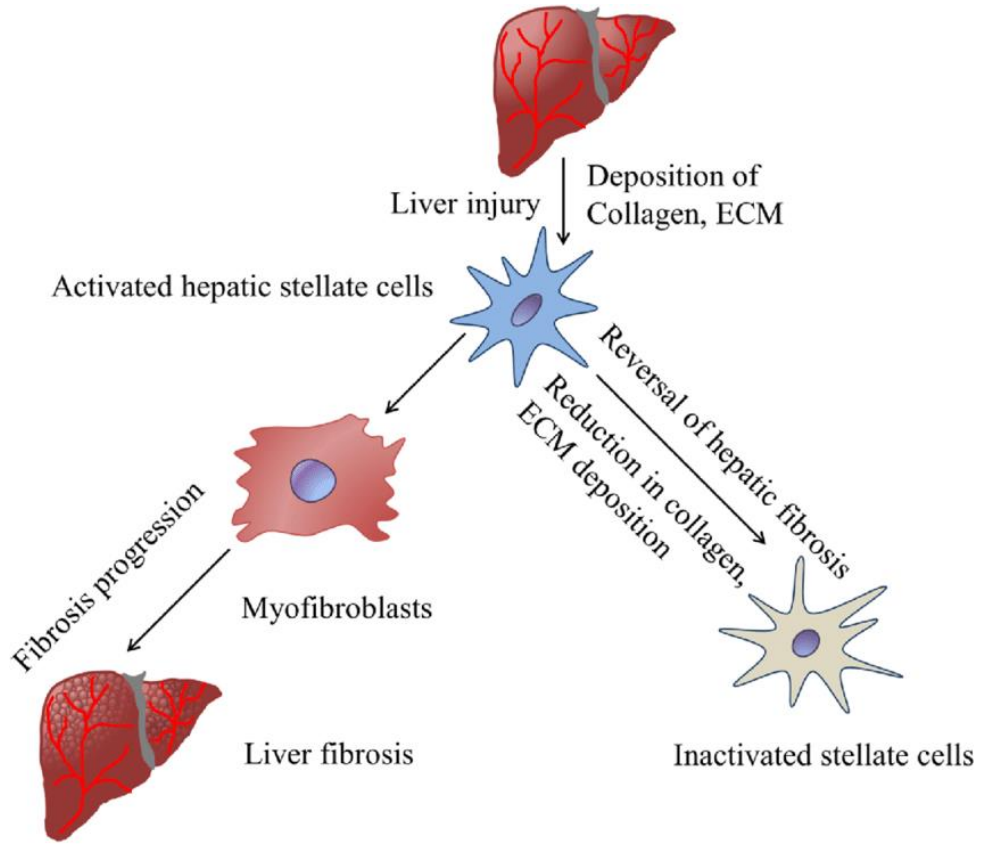


Key function: Immune defence and removal of unwanted material

Hepatic Stellate Cells

Also known as: Ito cells or lipocytes

- Located around the sinusoids
- Star-shaped cells with branching processes
- Store fat and retinoids, including vitamin A
- Help regulate vascular tone
- Can become activated during liver injury
- Major contributors to fibrosis and scar formation



inactive stellate cell → activated stellate cell → collagen / ECM deposition → fibrosis

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How Liver Cells Work Together

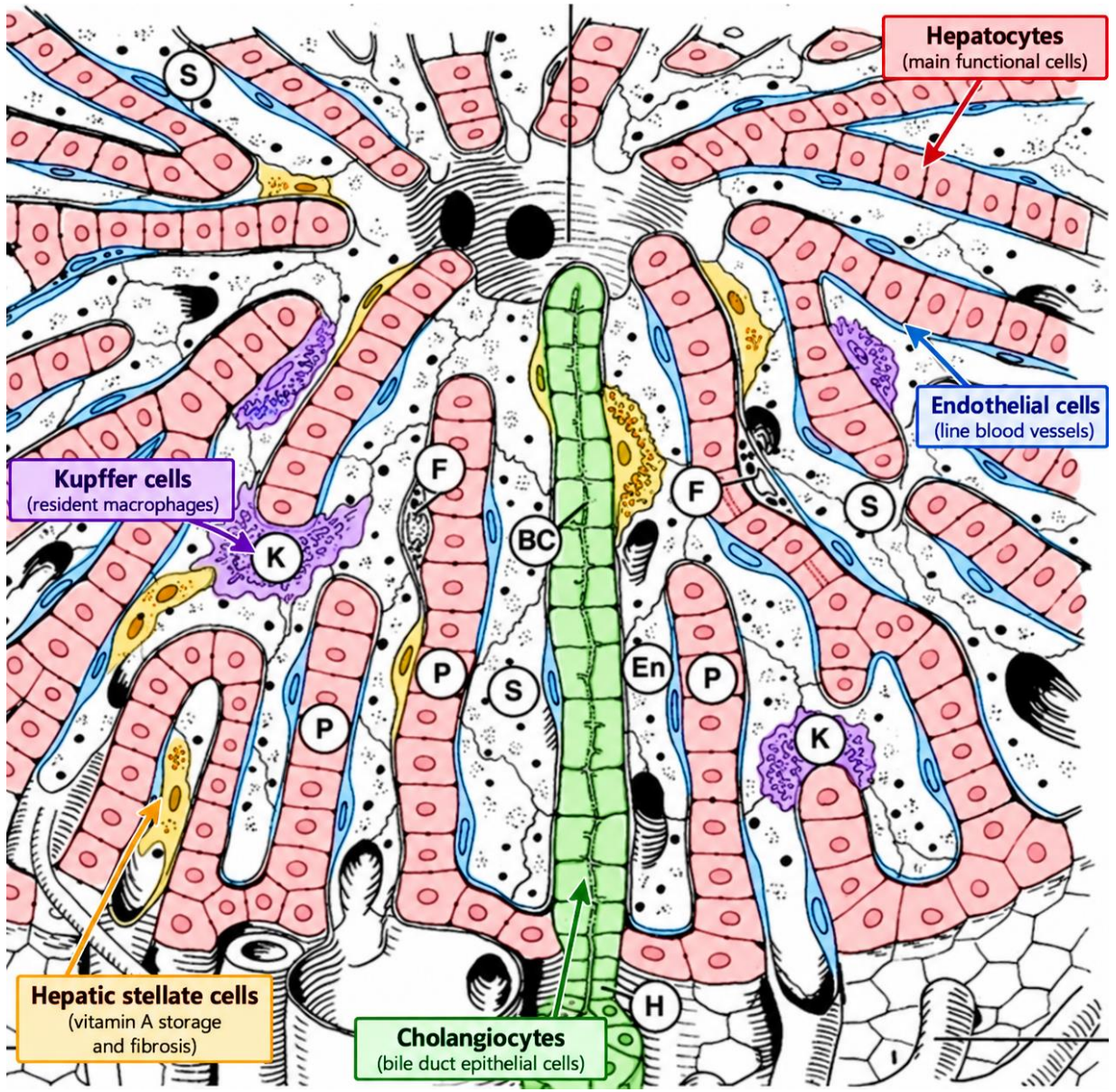
Hepatocytes
Metabolism and bile production

Cholangiocytes
Bile transport

Endothelial cells
Blood interface

Kupffer cells
Immune defence

Stellate cells
ECM and fibrosis



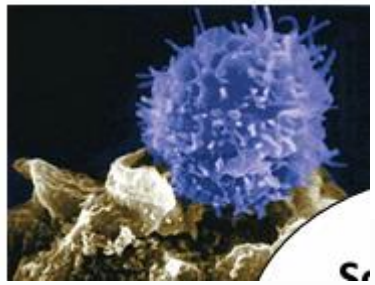
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Part 3 – Liver functions

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What Does the Liver Do?

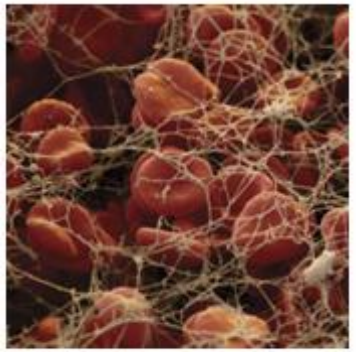




Five major functions of the liver

- Metabolism
- Detoxification
- Immune function
- Bile production
- Protein synthesis





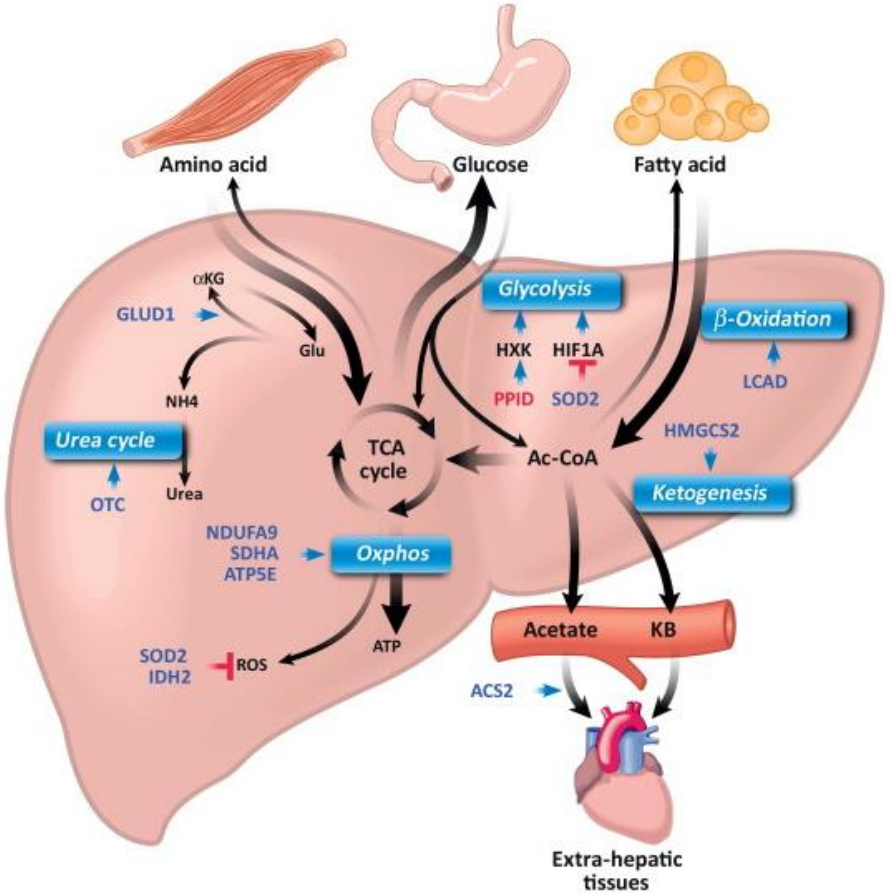


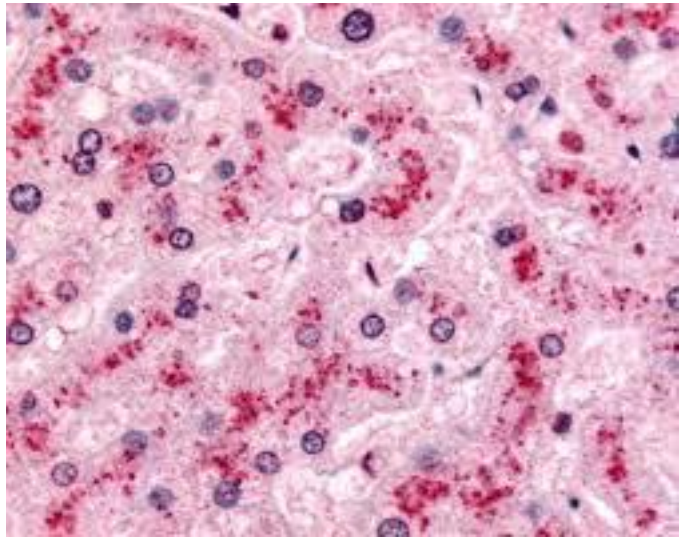


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Metabolic Functions of the Liver

- Carbohydrate metabolism (Glycogenolysis, Gluconeogenesis)
- Fat metabolism
- Protein metabolism
- Storage of vitamins and minerals





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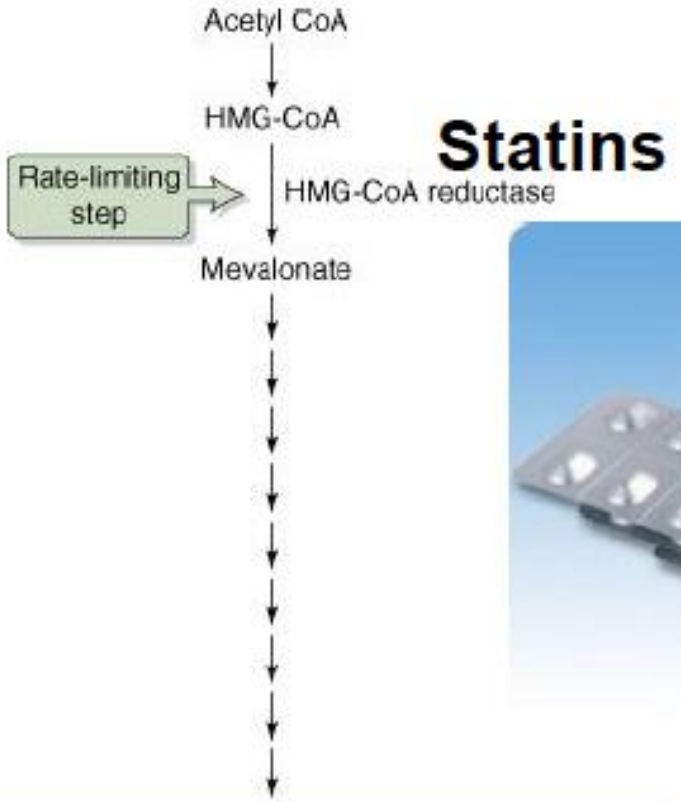
Cholesterol Synthesis

Cholesterol is important for:

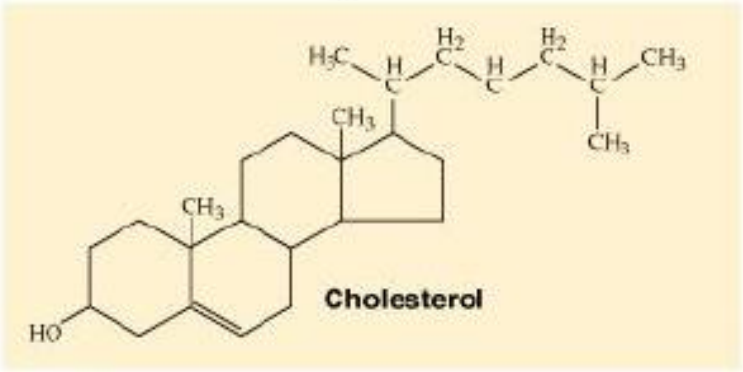
- Cell membrane structure and fluidity
- Bile acid production
- Steroid hormone synthesis
- Vitamin D production

Key enzyme: HMG-CoA reductase

Clinical connection: Statins







Annual global revenue

$19 billion

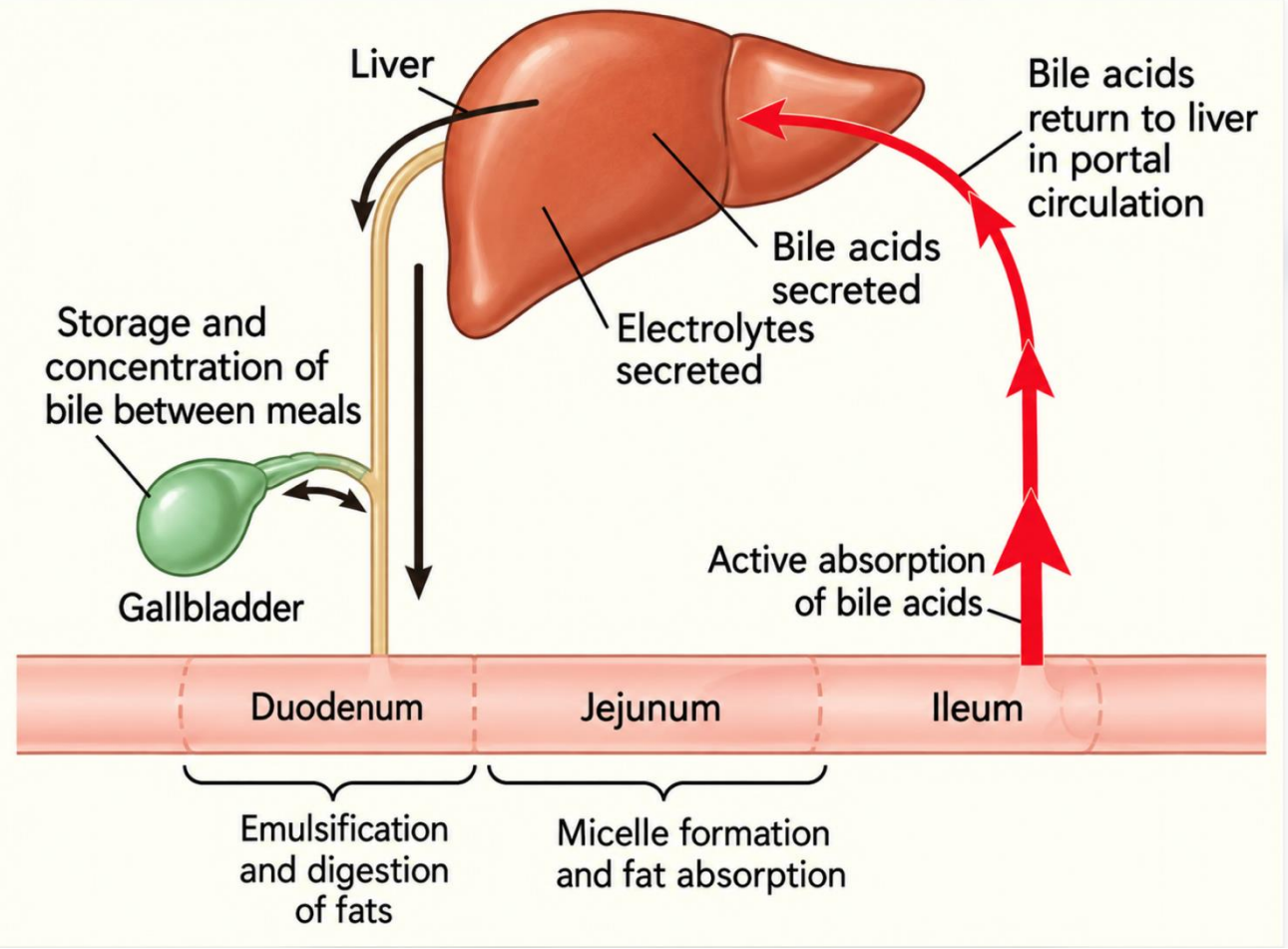
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Bile: Composition and Functions

Bile is produced by hepatocytes

Composition
- Water
- Bile salts
- Cholesterol
- Phospholipids
- Bilirubin

Functions
- Emulsification of fats
- Absorption of vitamins A, D, E and K
- Excretion of selected substances
- Intestinal immune regulation

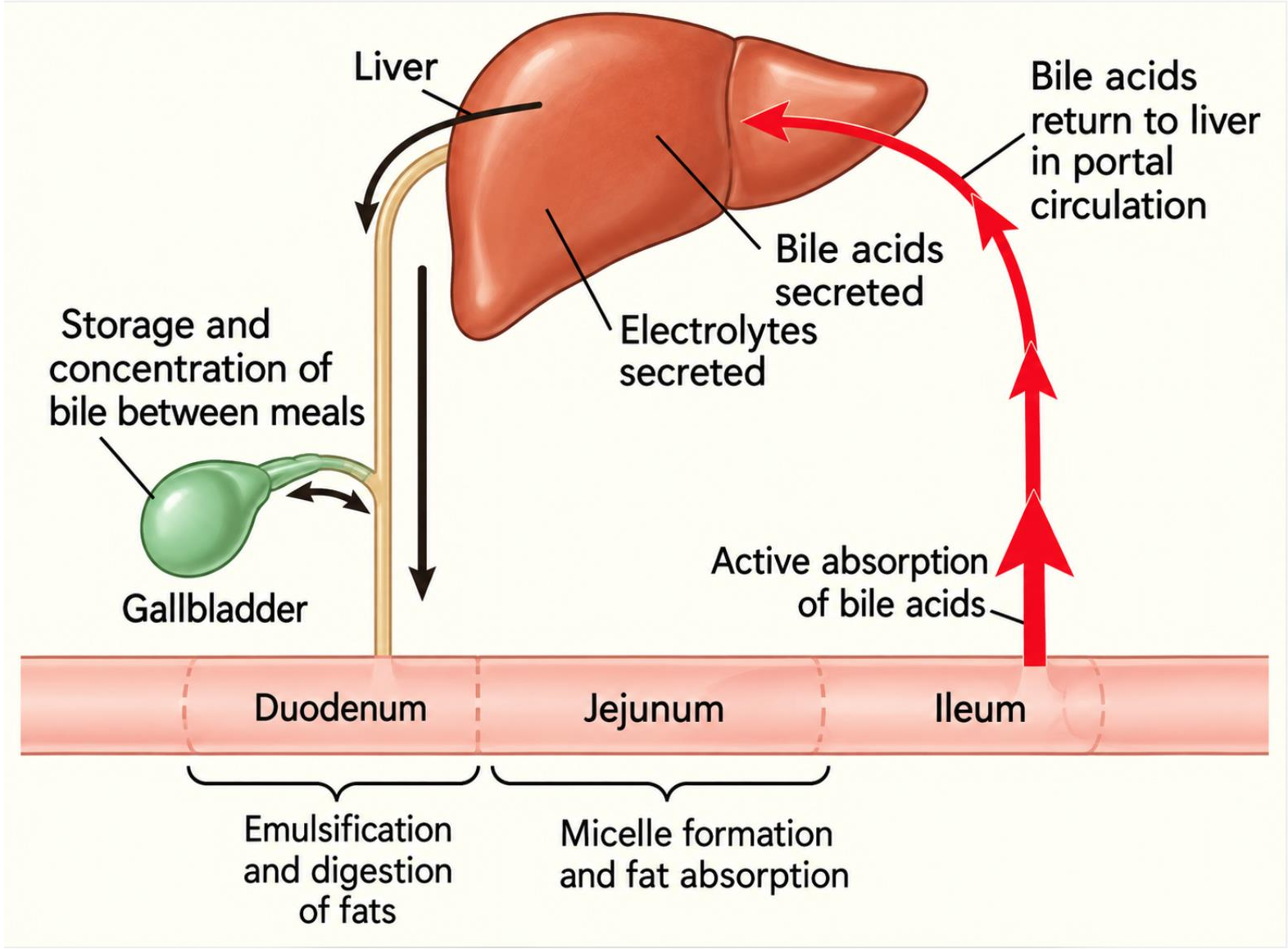


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Bile Production and Recycling

- Bile is continuously produced by hepatocytes.
- Bile is stored and concentrated in the gallbladder.
- Bile is released into the intestine when needed.
- Bile salts are recycled approximately 6–8 times per day.
- Recycling helps maintain the bile salt pool.

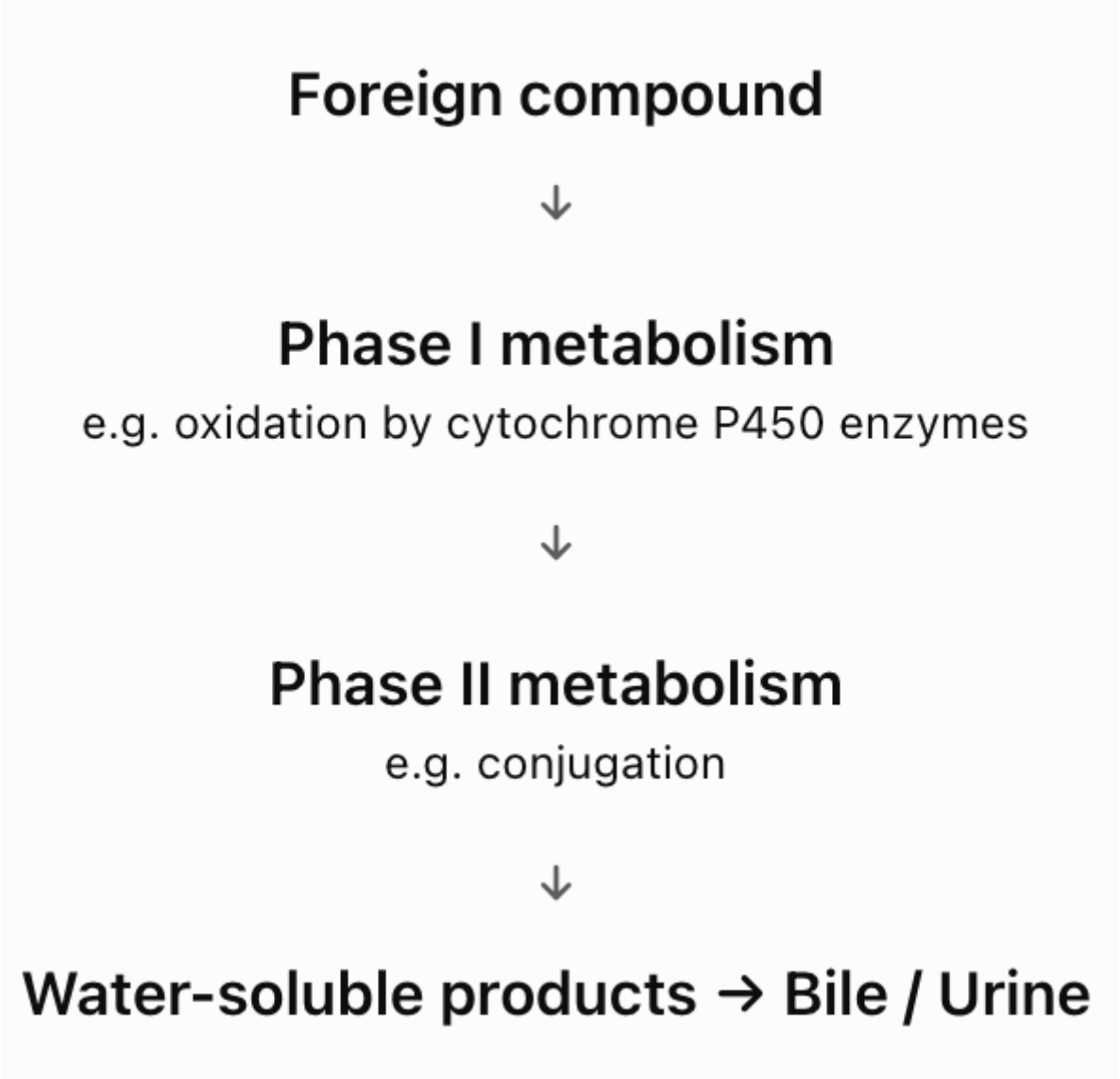
Liver → Gallbladder → Intestine → Liver



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Detoxification: Phase I and Phase II Metabolism

- **Cytochrome P450 (CYP)** : major enzymes involved in drug metabolism and deactivation, either directly or by facilitated excretion from the body.
- CYPs are also responsible for bioactivation of some compounds.
- Levels vary with age, gender, individual and cell/organ (or even area in organ)



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Synthetic Functions of the Liver



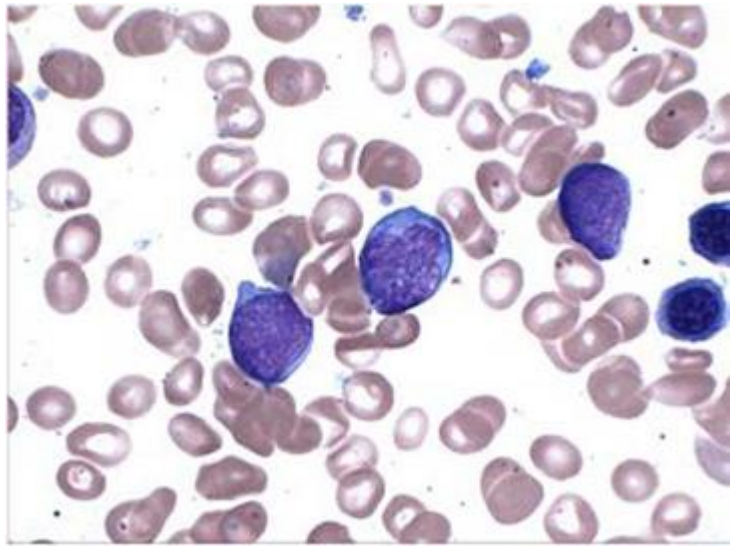
The liver produces important plasma proteins.

Protein	Main function
Albumin	Transport and osmotic regulation
Coagulation proteins	Blood clotting
Plasminogen	Coagulation-related processes
Alpha-1 antitrypsin	Inflammation regulation
Transferrin	Iron transport
Hepcidin	Iron regulation

Immune Functions of the Liver

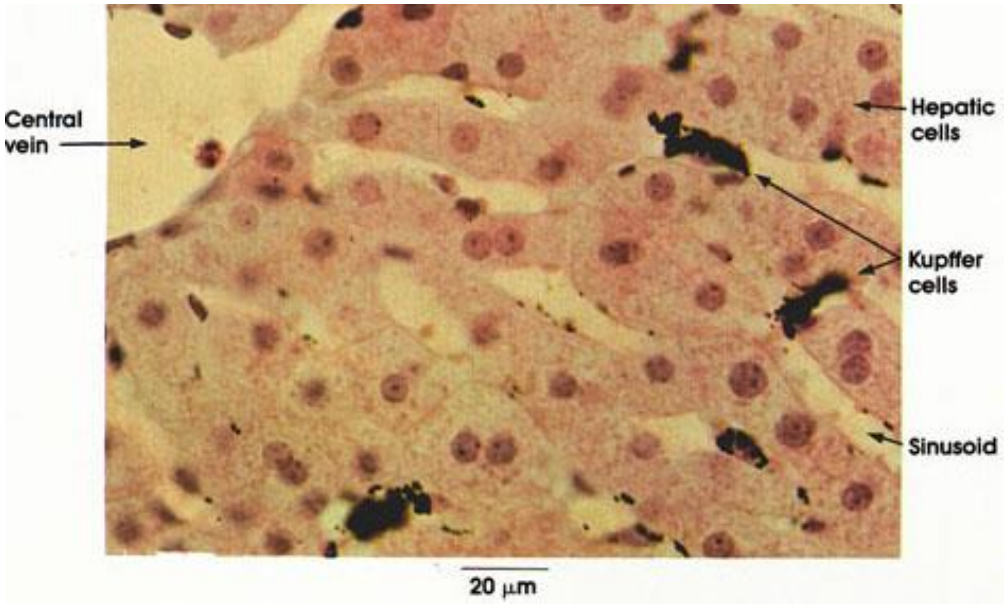
The liver contributes to immune protection.

- Protection against blood-borne pathogens
- Phagocytosis of old or dying cells
- Innate immune responses
- Induction of immune tolerance



Important immune components

- Kupffer cells
- Natural killer (NK) cells
- Lymphocytes
- Acute-phase proteins

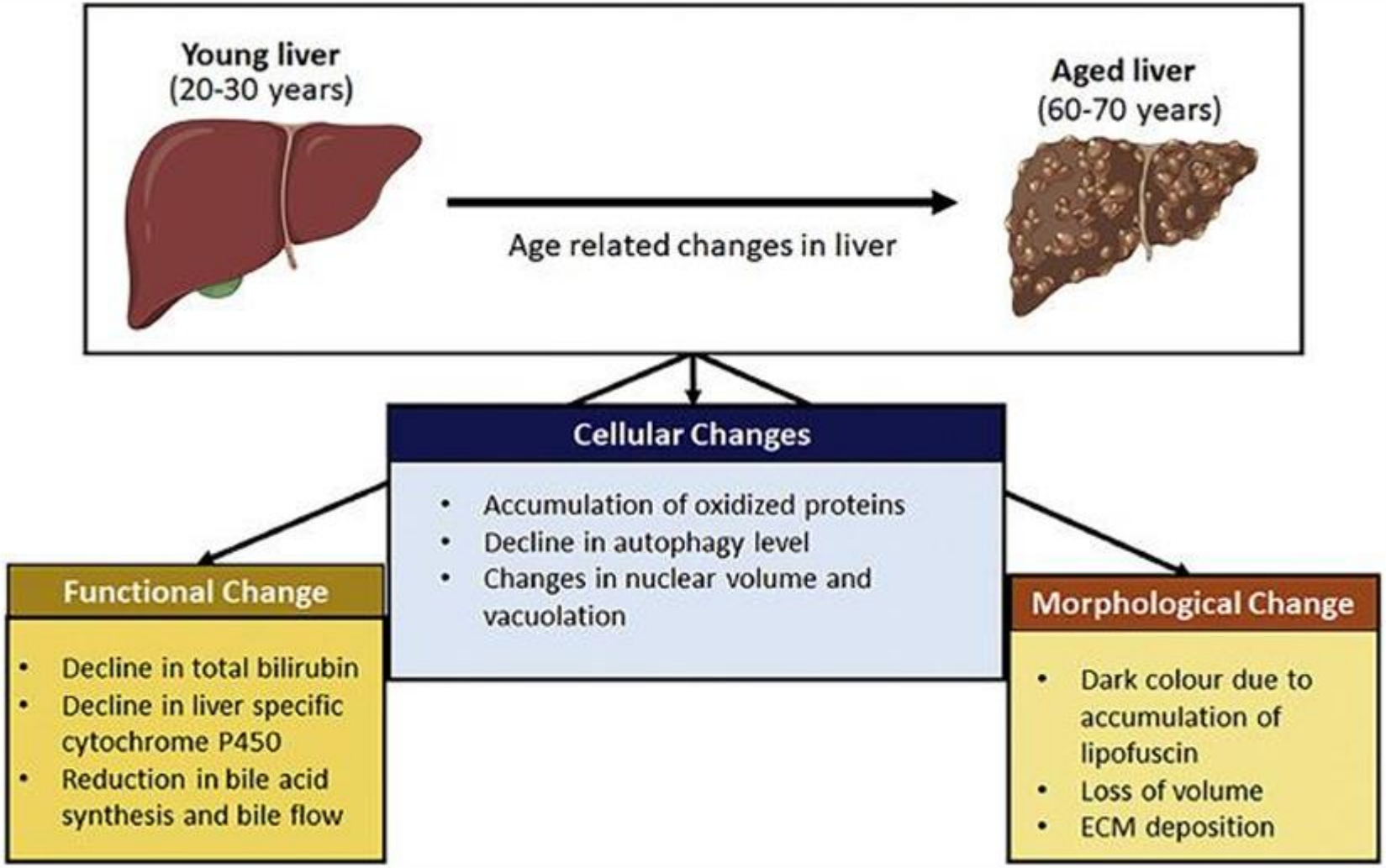


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Liver Function Changes with Age

Age-related changes may affect:

- Liver volume and morphology
- Cellular protein accumulation
- Autophagy
- Cytochrome P450 activity
- Bile acid synthesis and bile flow
- Overall detoxification capacity



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Part 4 – Liver diseases

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Why Do We Care About Liver Disease?

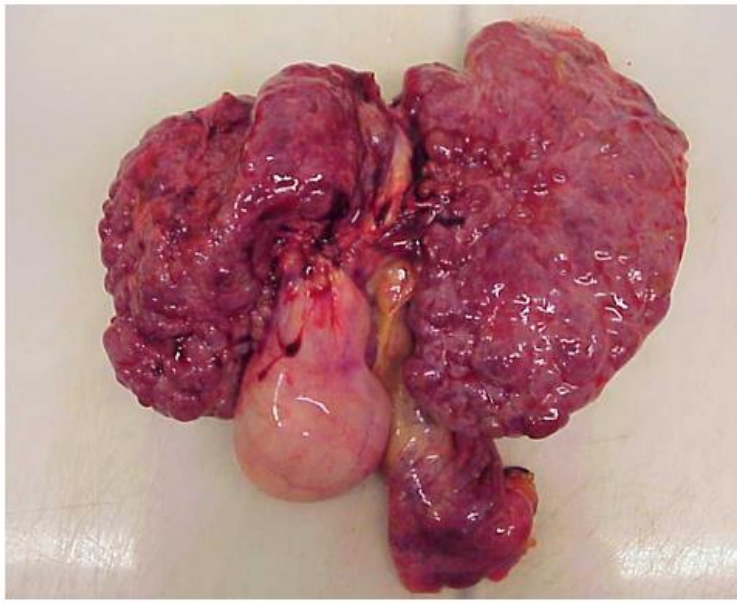
- Chronic liver disease is a major health problem.
- Liver injury may progress to fibrosis and cirrhosis.
- Cirrhosis can cause life-threatening complications.
- Liver disease has multiple causes.

Major Causes of Liver Disease

Viral infection	Alcohol	Metabolic factors
Hepatitis viruses	Alcohol-related injury	MASLD

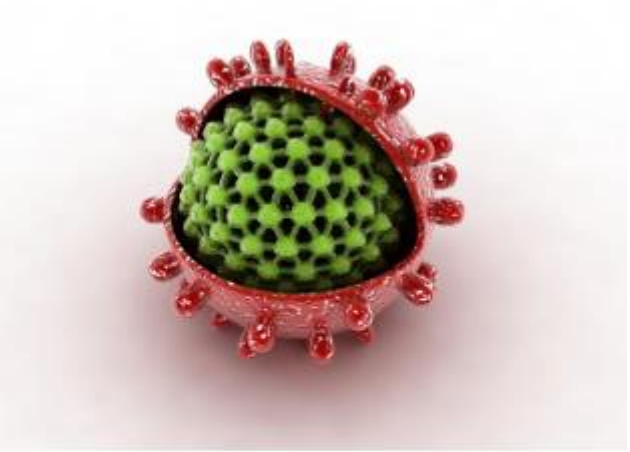
Other causes include toxins and autoimmune injury.





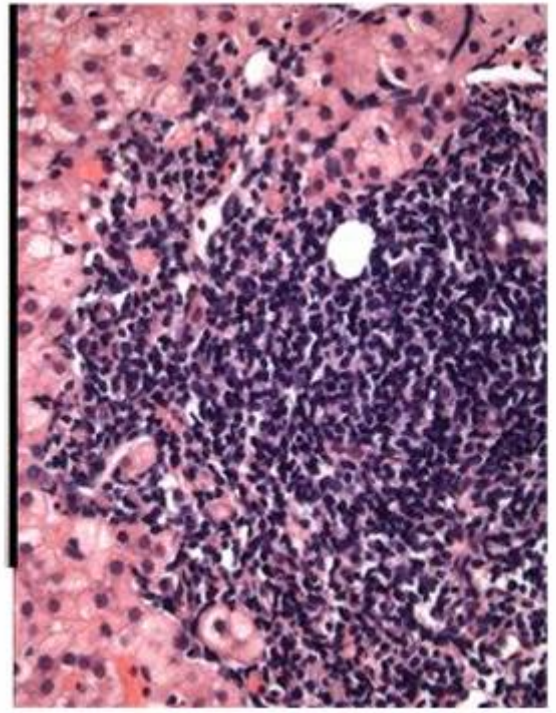
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Viral Hepatitis



- Hepatitis viruses A–E can infect the liver.
- Immune responses contribute to hepatocyte injury.
- Some infections resolve.
- Others may become chronic.
- Chronic inflammation can lead to fibrosis and liver failure.

HBV and HCV are important causes of chronic viral hepatitis.



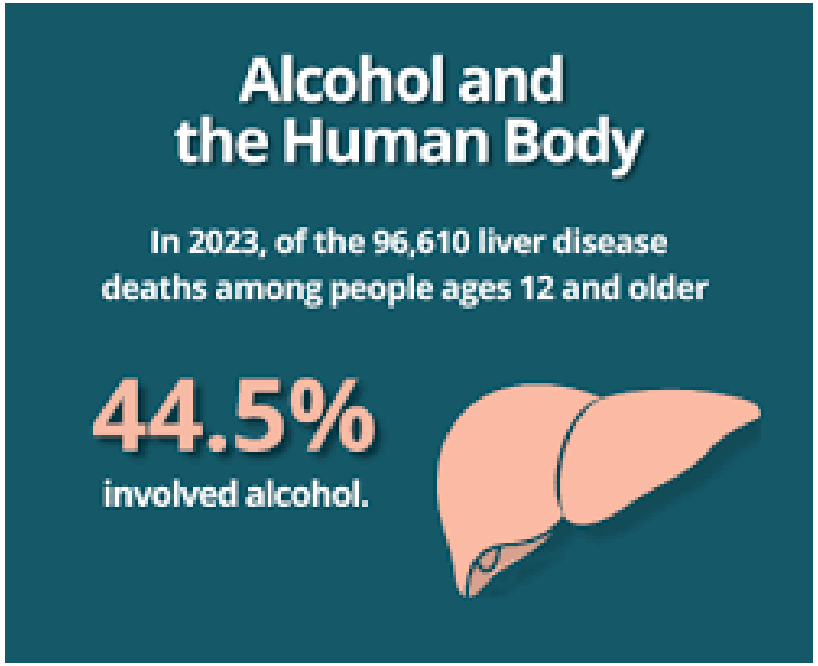
Viral infection → Immune response → Hepatocyte injury → Chronic inflammation → Fibrosis

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Alcohol-Related Liver Disease



- Alcohol is a major cause of liver disease.
- Excessive alcohol consumption can damage liver tissue.
- Alcohol-related injury may contribute to cirrhosis.
- Alcohol and metabolic factors can both contribute to liver disease.



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Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD)

MASLD

- Fat accumulation in more than 5% of hepatocytes
- Associated with:
  Obesity
  Diabetes
  Dyslipidaemia
  Hypertension
- May progress to inflammation and fibrosis
- Increased risk of hepatocellular carcinoma (HCC)

Obesity
Diabetes
Dyslipidaemia
hypertension



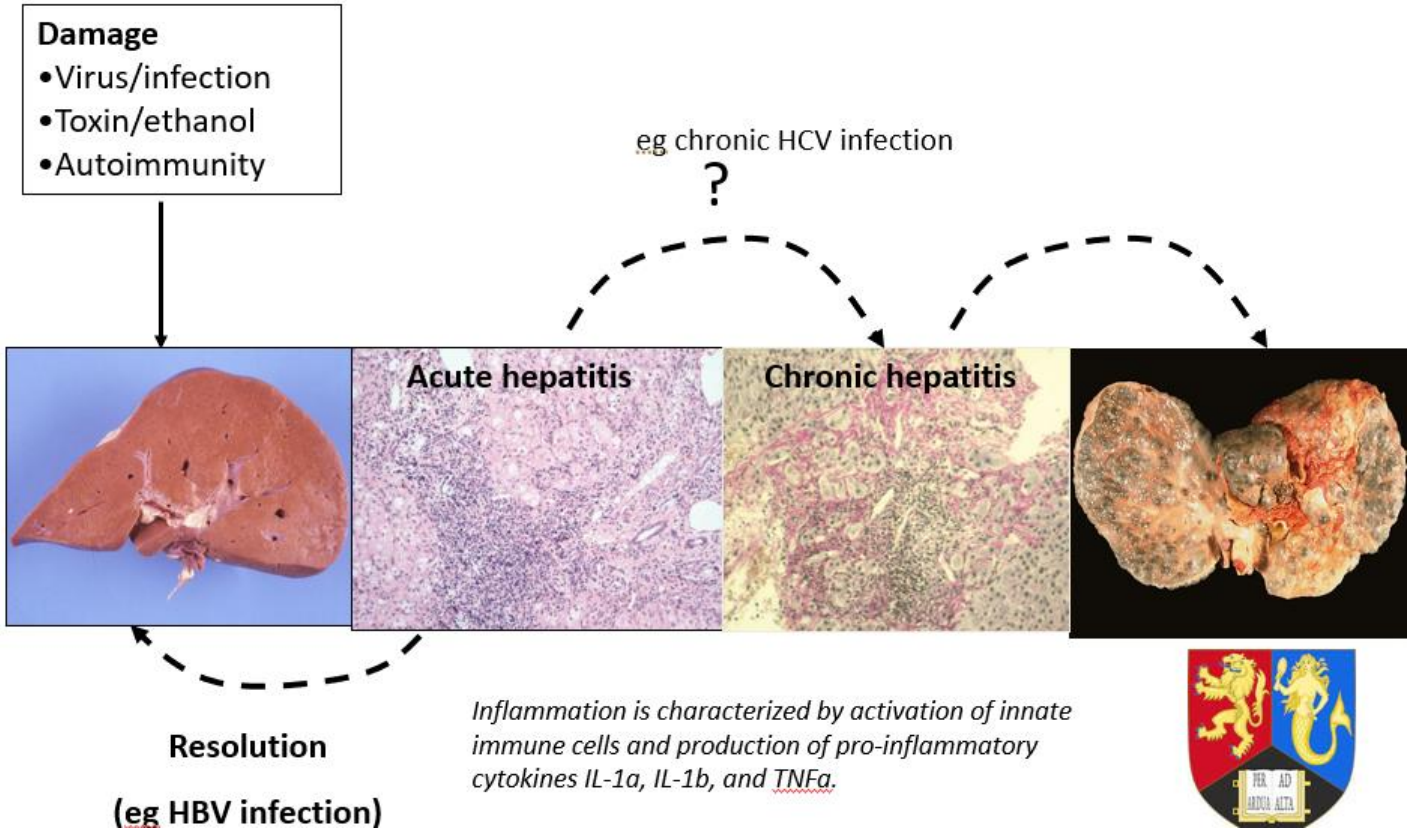
Insulin resistance
Lipotoxicity
Oxidant stress
Apoptosis
inflammation

Fat accumulation → Inflammation → Fibrosis



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Different Causes, Common Pathways



Liver injury

Virus / Toxins / Alcohol / Autoimmunity

↓

Acute hepatitis

↙ Resolution

↘ Chronic hepatitis

↓

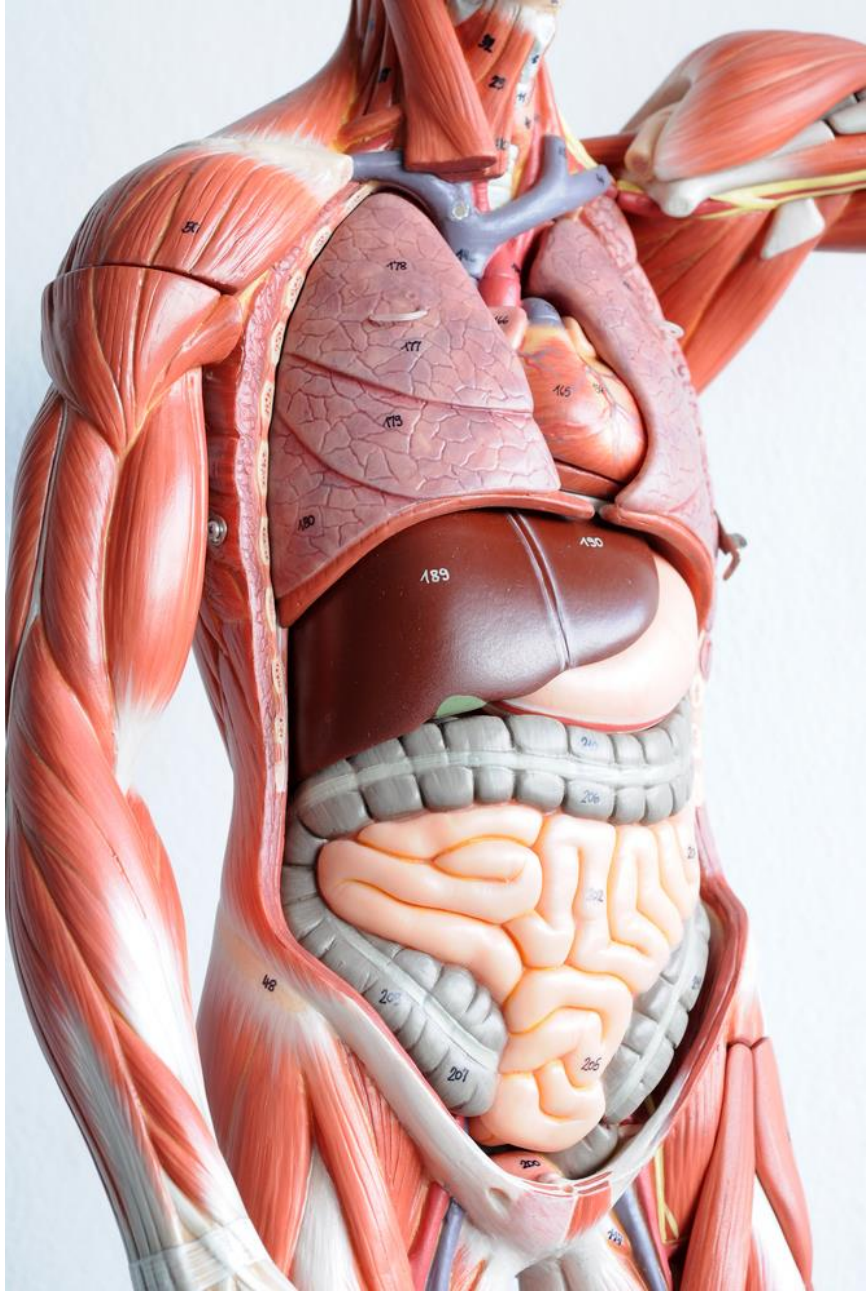
Fibrosis / Cirrhosis

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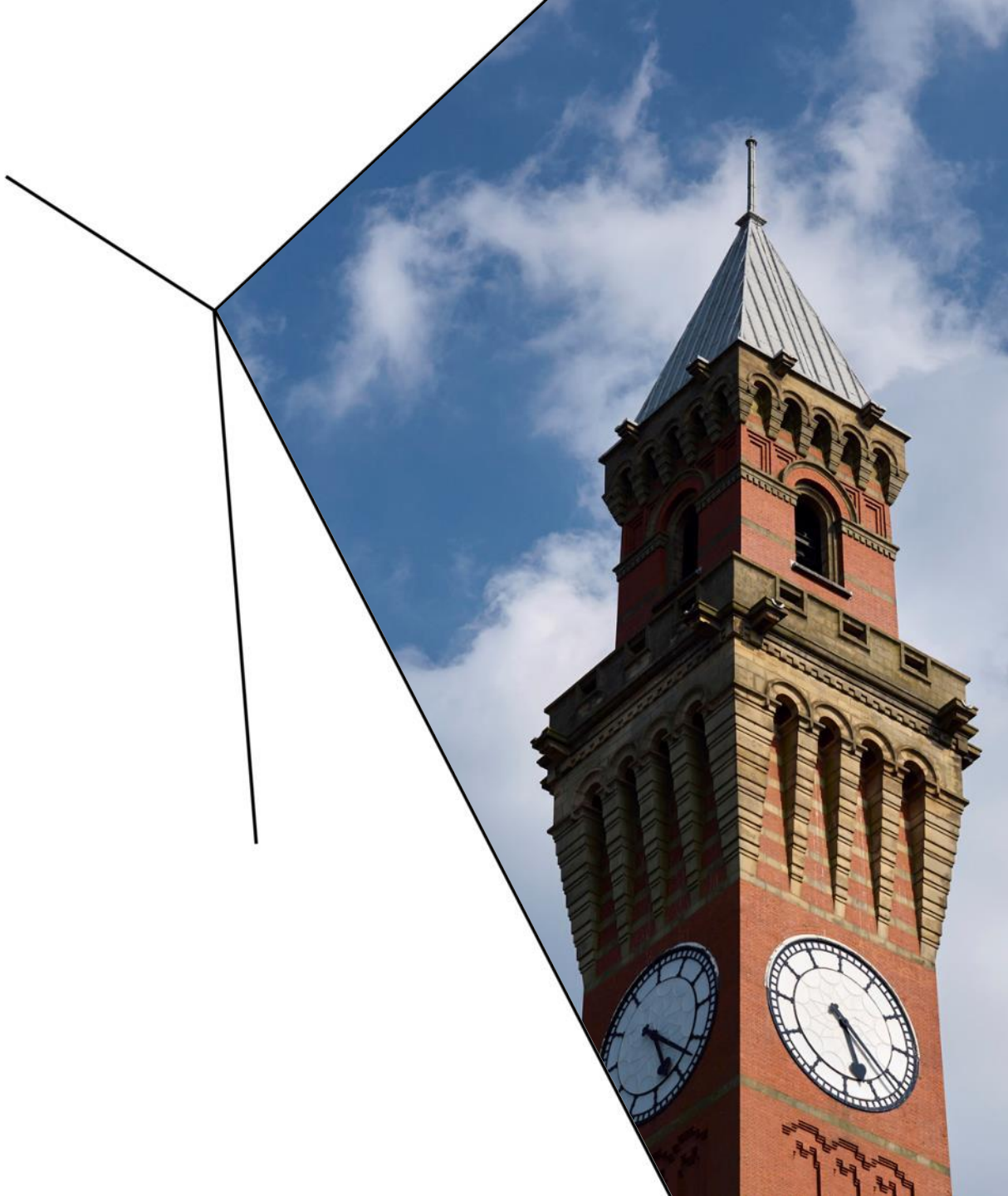
Take-Home Messages

By the end of this lecture, you should remember:

1. The liver has a complex gross and microscopic structure.
2. Five major hepatic cell types have specialised functions.
3. The liver performs metabolic, synthetic, biliary, detoxification and immune functions.
4. Liver disease has multiple causes.
5. Chronic injury can lead to fibrosis and cirrhosis.



Questions



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